

ANATOMY OF THE ARM

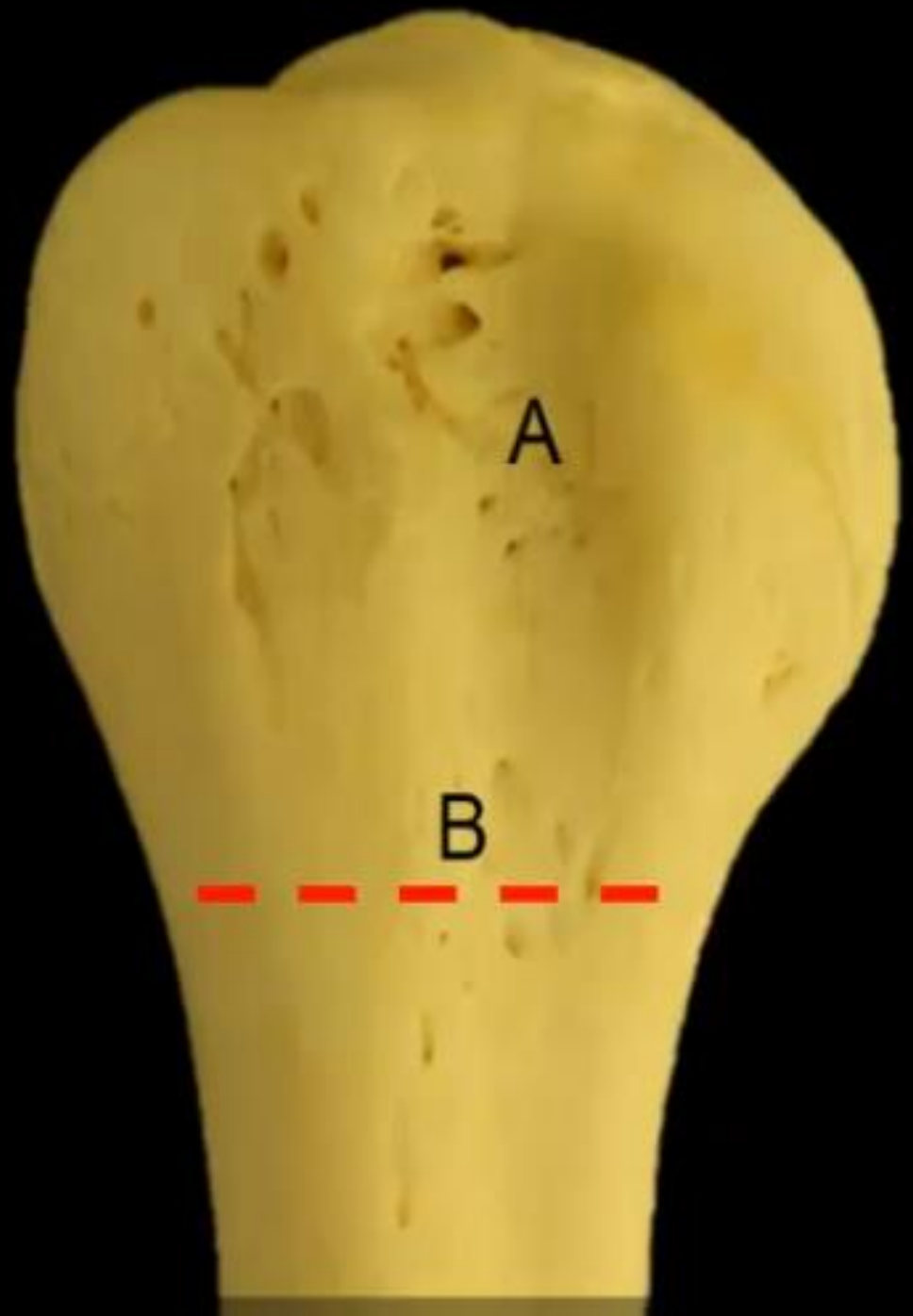
DR. AHMED AL-BAGHDADI, *Ph.D*

THE ARM

- Bone
- Fascia
- Anterior compartment
 - Biceps brachii
 - Brachialis
 - Coracobrachialis
- Posterior compartment
 - Triceps
- ARTERIES & SUPERFICIAL VEINS
 - Axillary artery
 - Anastomosis around the scapula
 - Brachial artery
 - Anastomosis around the elbow
- Elbow joint
- proximal radio-ulnar joint
- Cubital fossa

BONE





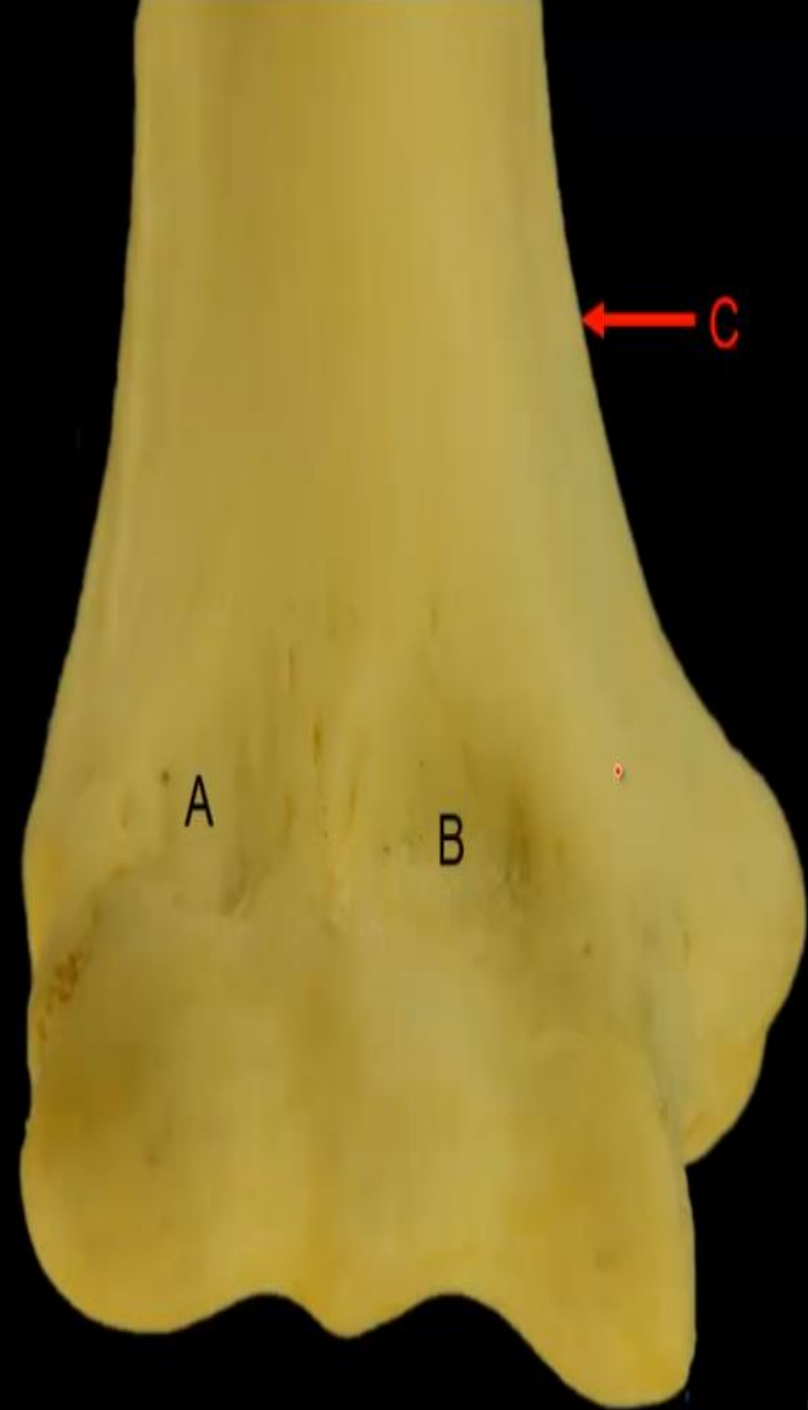
Lateral epicondyle

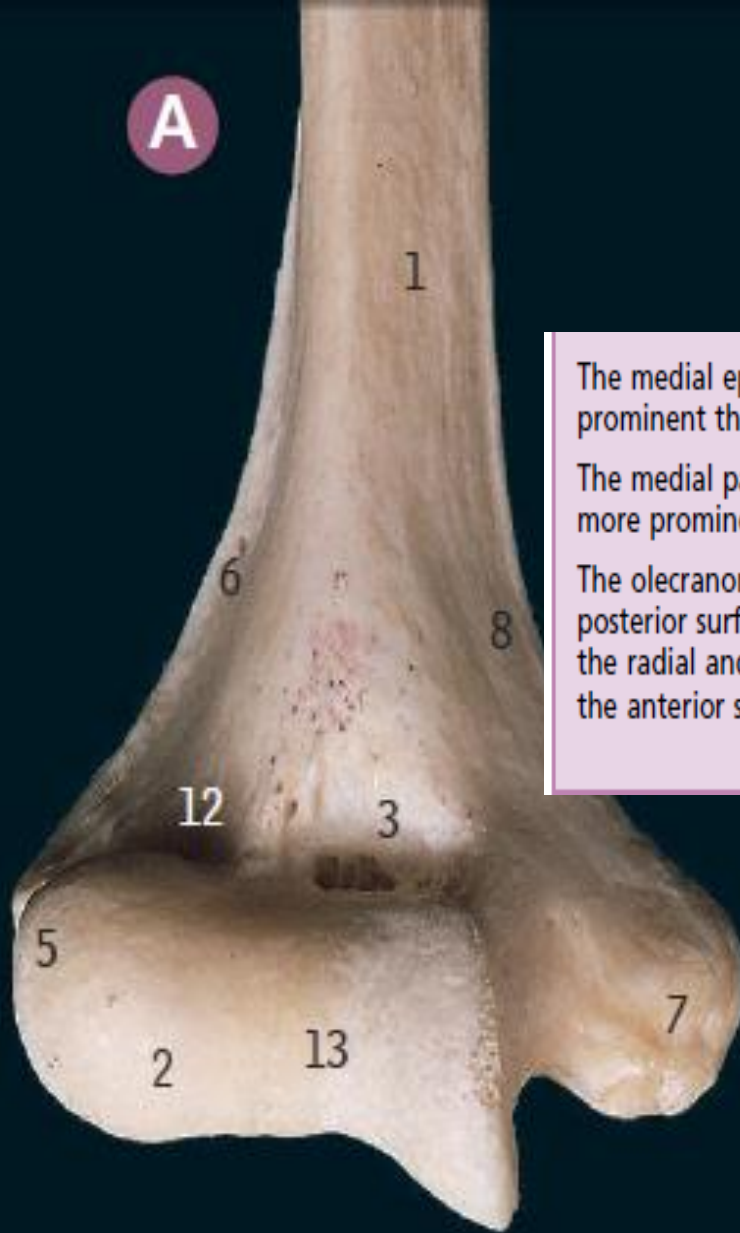


Medial epicondyle

Golfers elbow

©MMG 2000



A**B**

The medial epicondyle (7) is more prominent than the lateral (5).

The medial part of the trochlea (13) is more prominent than the lateral part.

The olecranon fossa (10) on the posterior surface is deeper than the radial and coronoid fossae on the anterior surface (12 and 3).



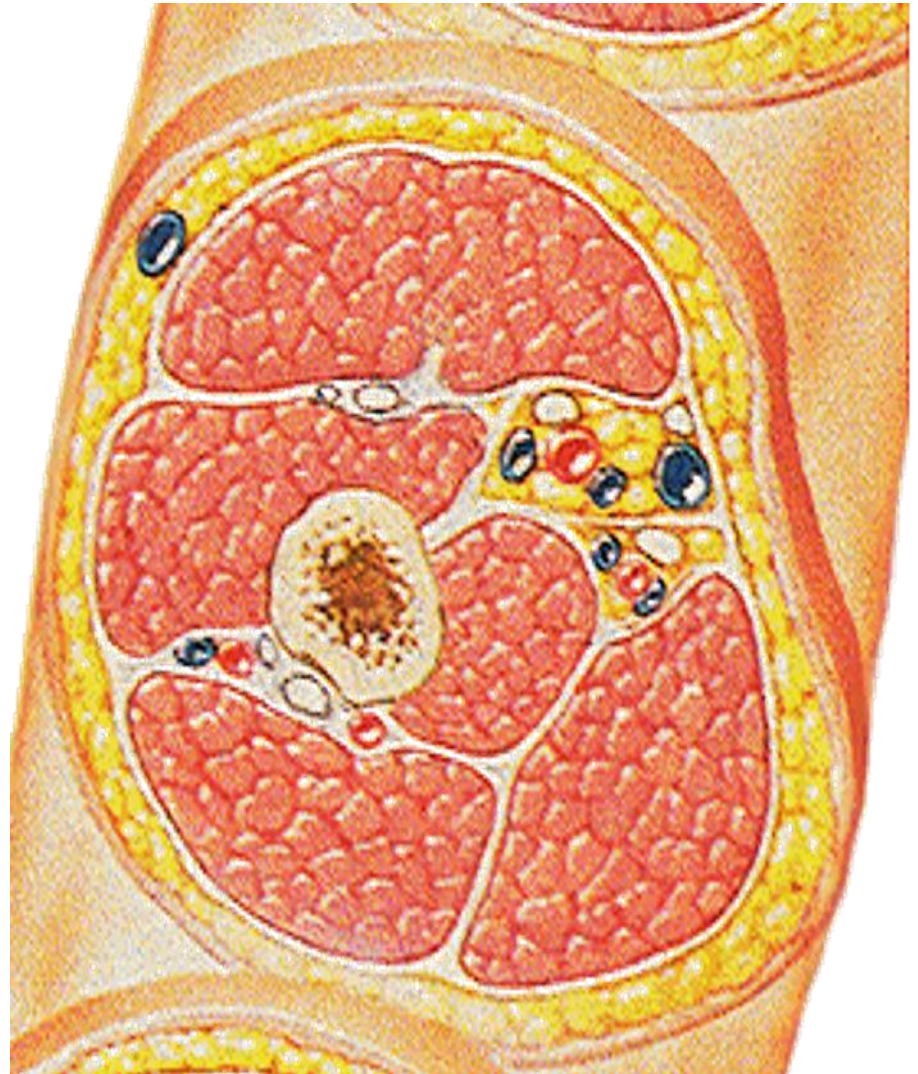


The trochlear notch (11) faces forwards, with the radial notch (9) on the lateral side.

FASCIA

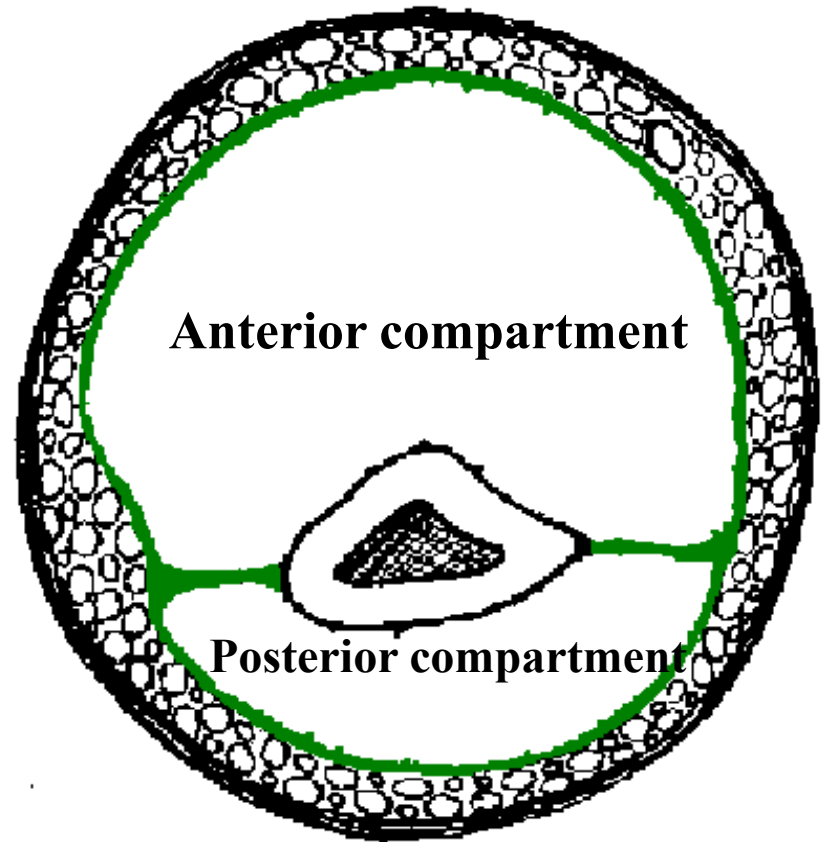
Deep Fascia

- The arm is enclosed in a sheath of deep fascia. from which
- two fascial intermuscular septa extend to the medial and lateral supracondylar ridges of the humerus



Compartments

- The medial and lateral intermuscular septa divide the arm into
- anterior and posterior fascial compartments.



COMPARTMENTS

ANTERIOR COMPARTMENT

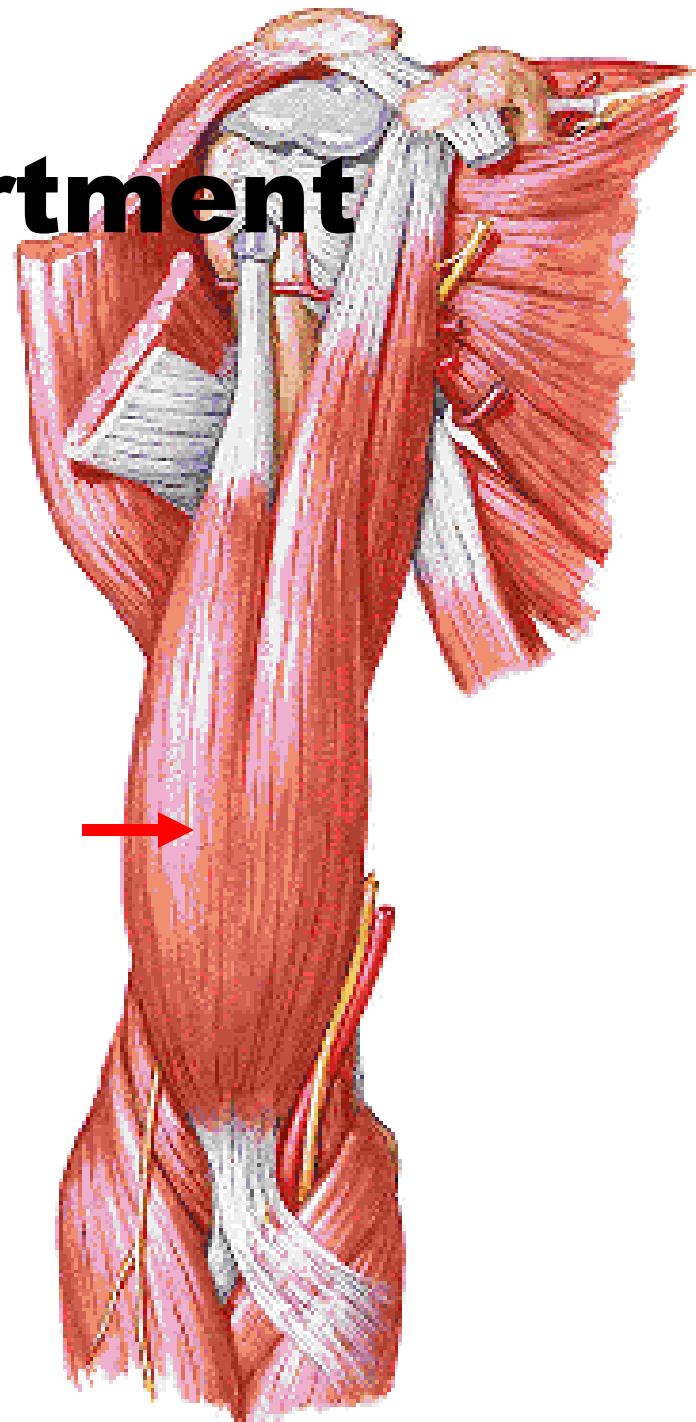
A. Muscles of front of arm

Muscle	Origin	Insertion	Nerve supply	Action.
1-Biceps Brachii	a. Long head: supraglenoid tubercle (intra-capsular extra-synovial). b. Short head: coracoid process.	Posterior part of radial tuberosity & deep fascia of upper part of medial side of forearm.	All muscles of front of arm are supplied by musculo-cutaneous nerve except lateral ½ of	Flexion of supinated forearm. Supination of flexed pronated forearm at radioulnar joints. • Weak flexor of shoulder.
2- Coracobrachialis	Coracoid process (with short head of biceps)	Middle of medial surface of humerus	brachialis (supplied by radial nerve i.e. double	Flexion & adduction of shoulder.
3- Brachialis	Lower 1/ 2 of humerus	Ulnar tuberosity.	nerve supply)	Main flexor of elbow.

Anterior compartment

contains **THREE** muscles

□ Biceps brachii

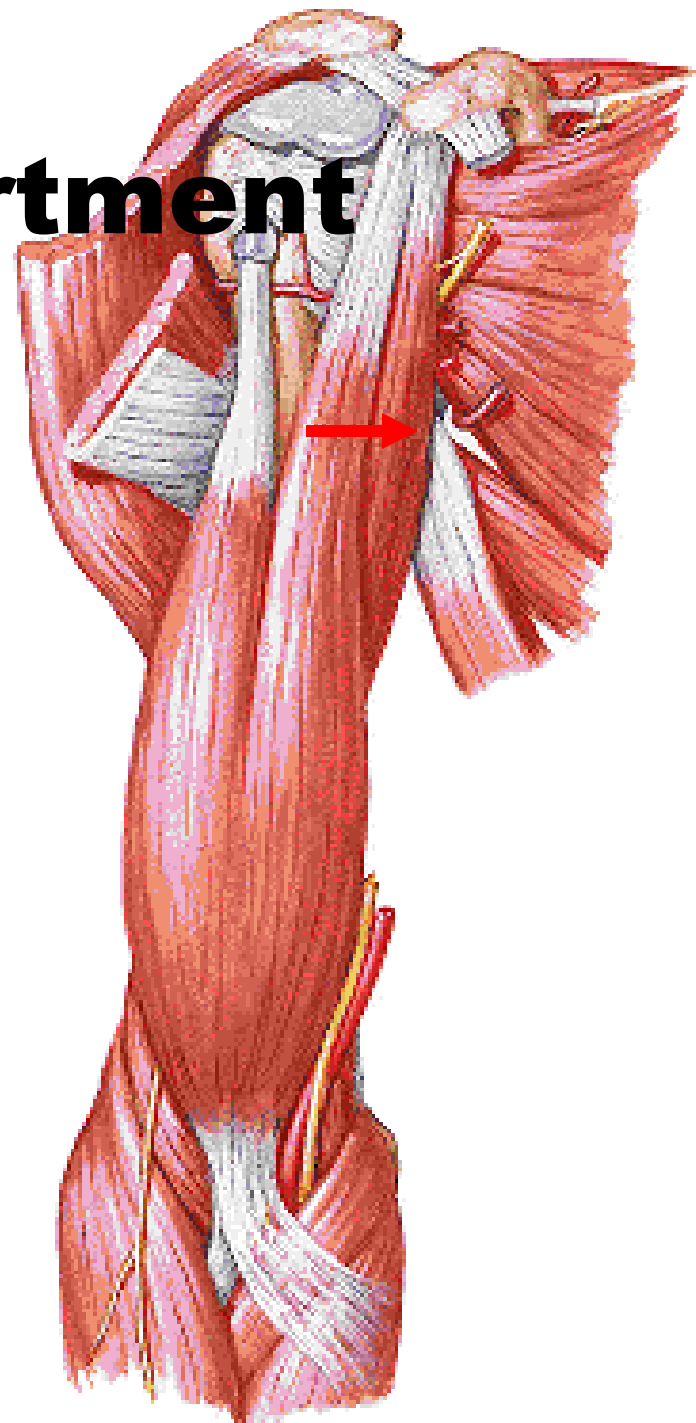


Anterior compartment

contains **THREE** muscles

□ Biceps brachii

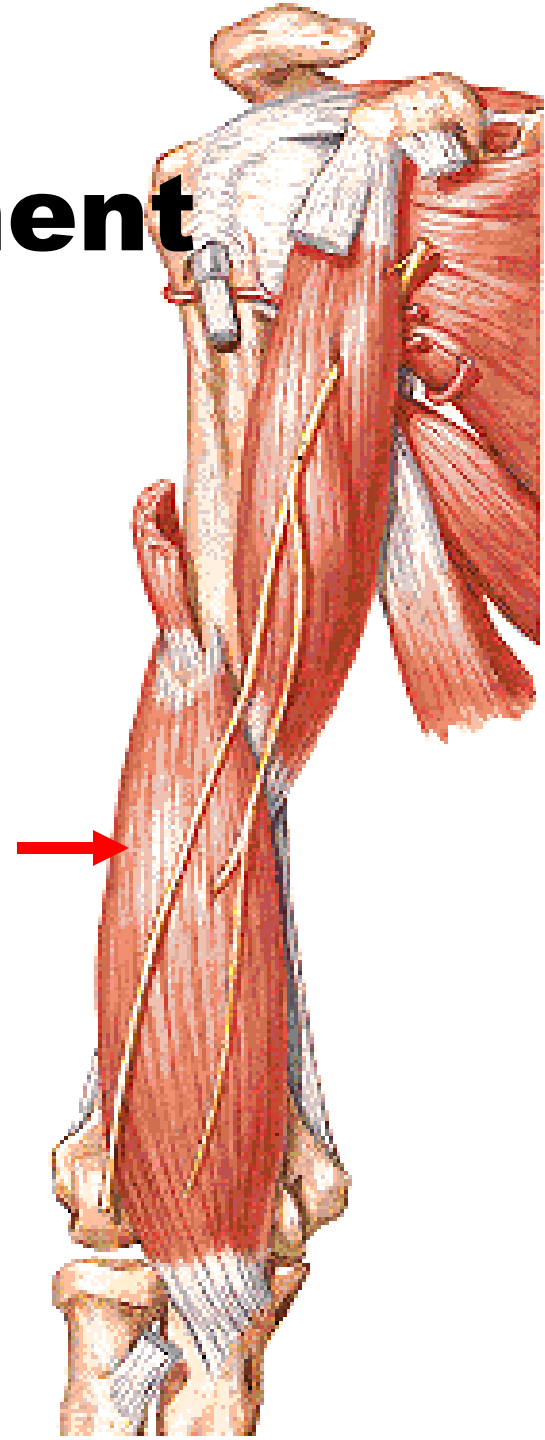
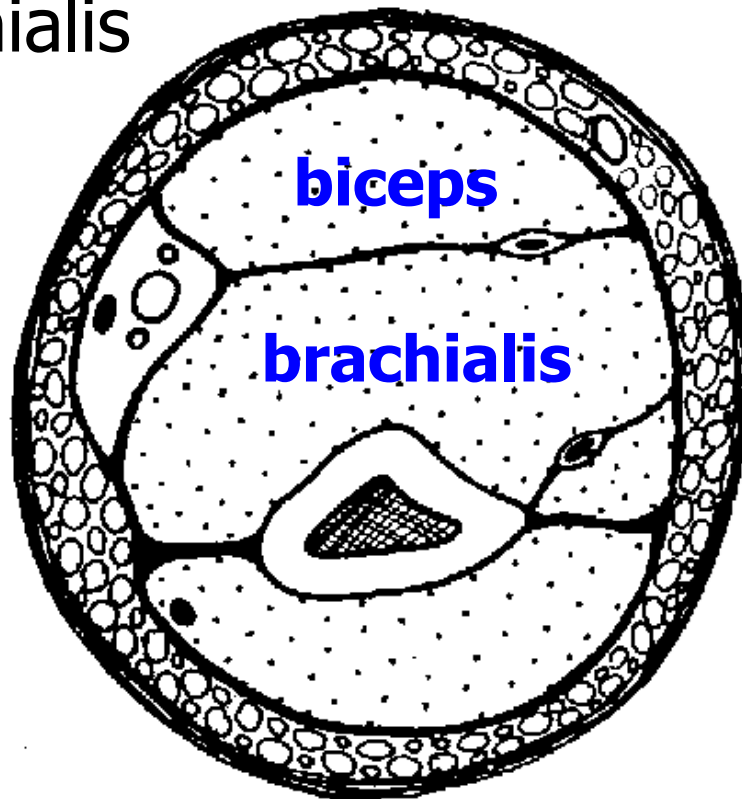
□ Coracobrachialis



Anterior compartment

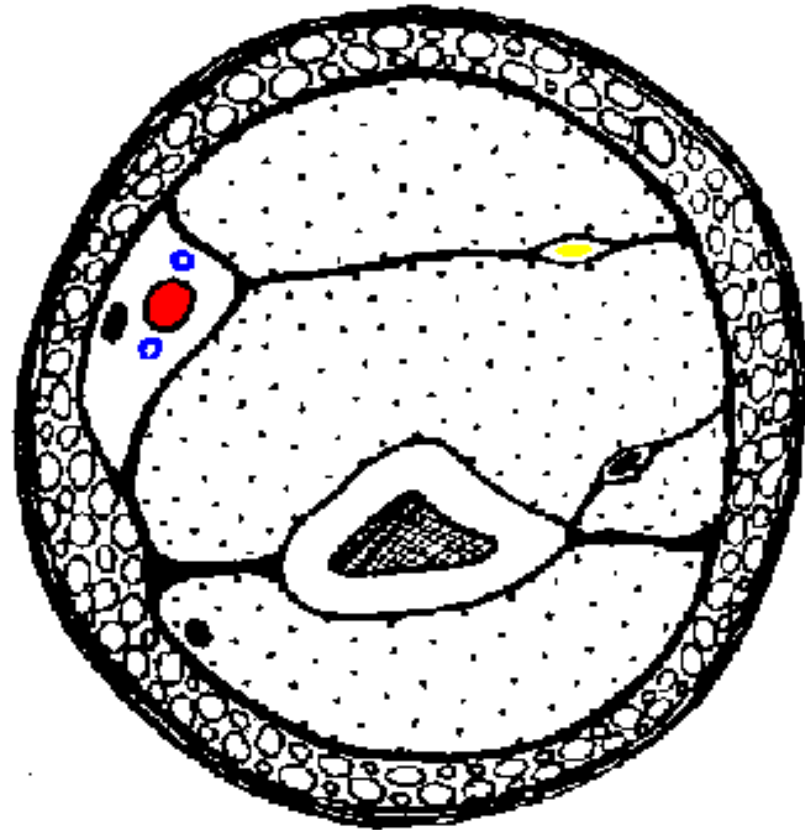
contains **THREE** muscles

- Biceps brachii
- Coracobrachialis
- Brachialis



Anterior compartment

- nerve of the compartment is
musculocutaneous nerve
- artery of compartment is
brachial artery



Biceps brachii

□ Origin

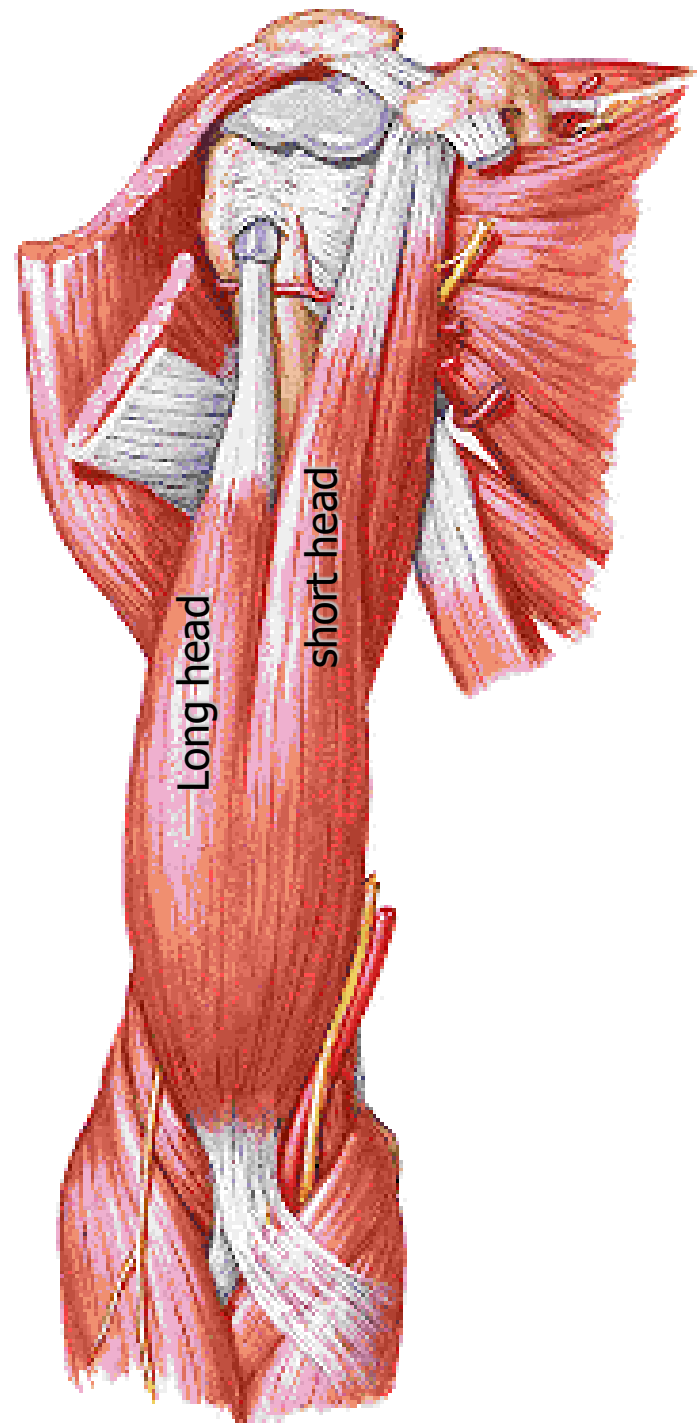
has 2 heads (as its name implies)

- long head

(which is lateral, L. for L.)

- short head

which is medial



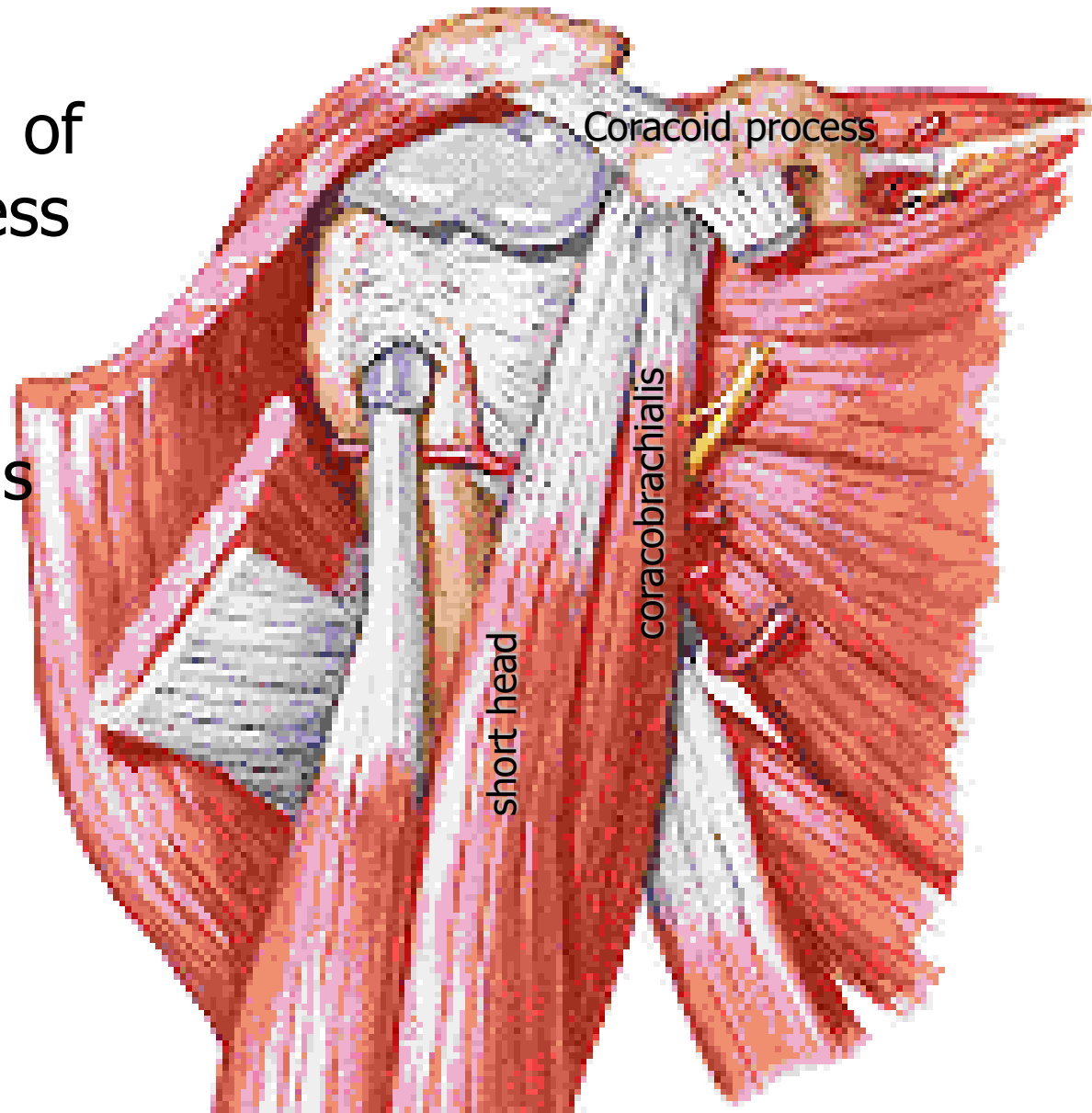
Long head of biceps brachii



- emerges at the top of the intertubercular sulcus
- surrounded by a tube-like extension of synovial membrane

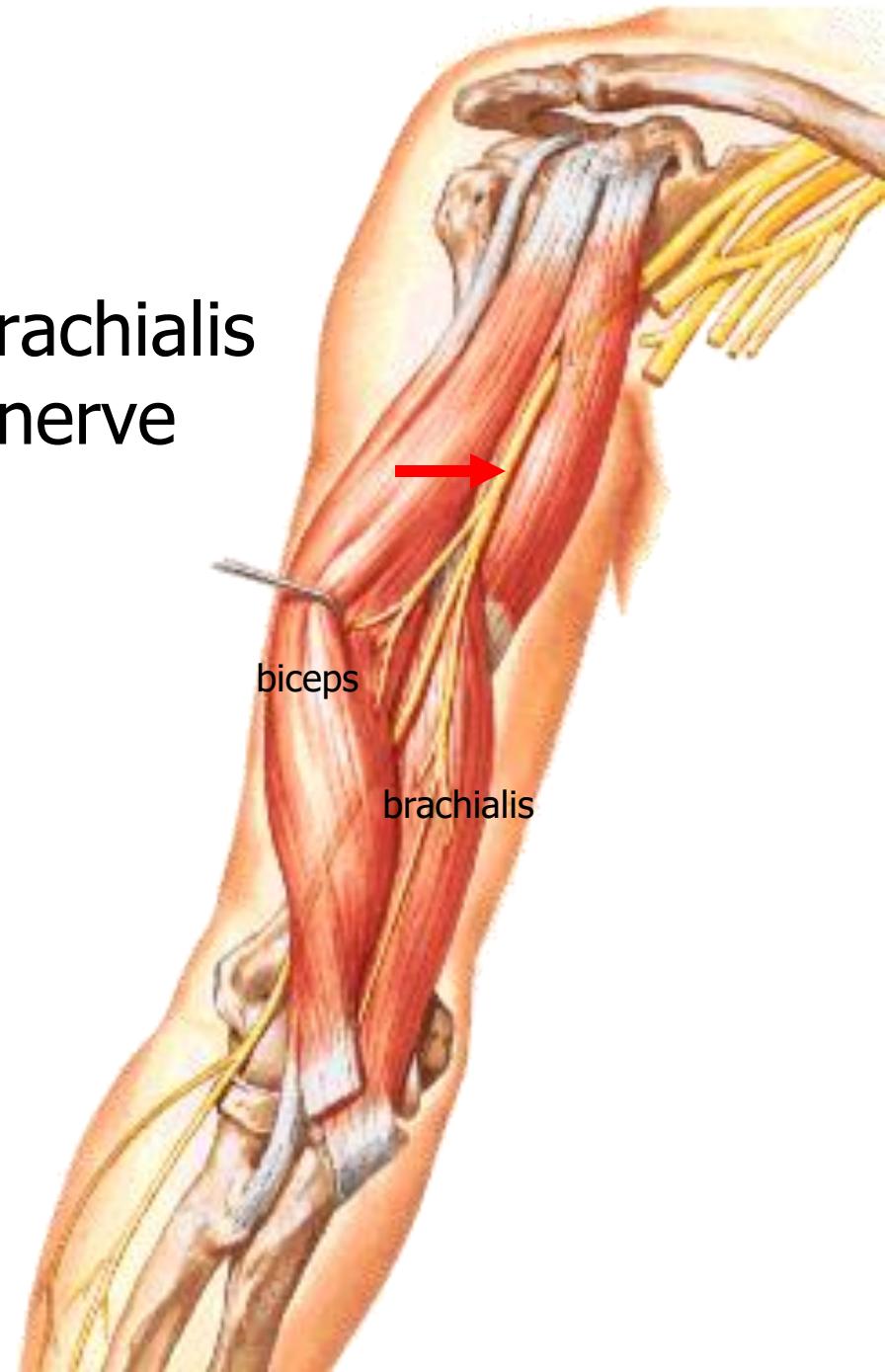
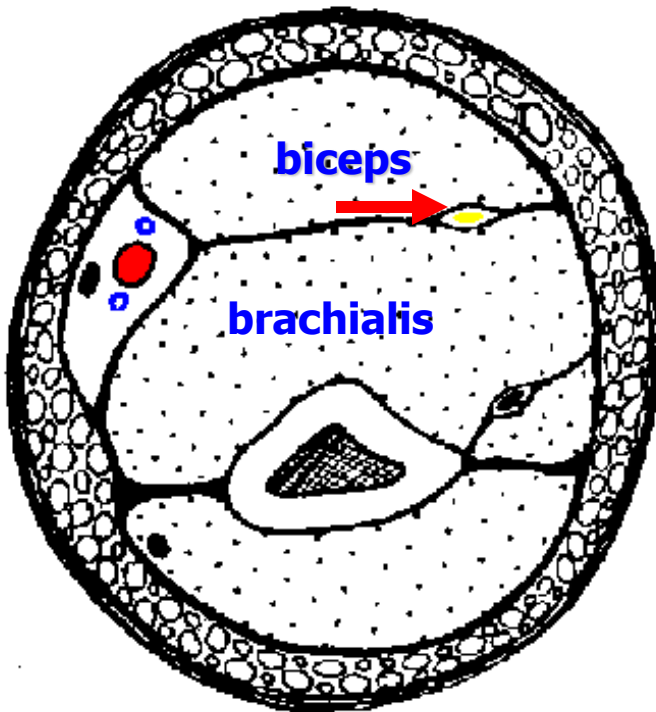
Medial head of biceps brachii

□ arises from the tip of the coracoid process of the scapula in common tendon with coracobrachialis muscle



Biceps brachii

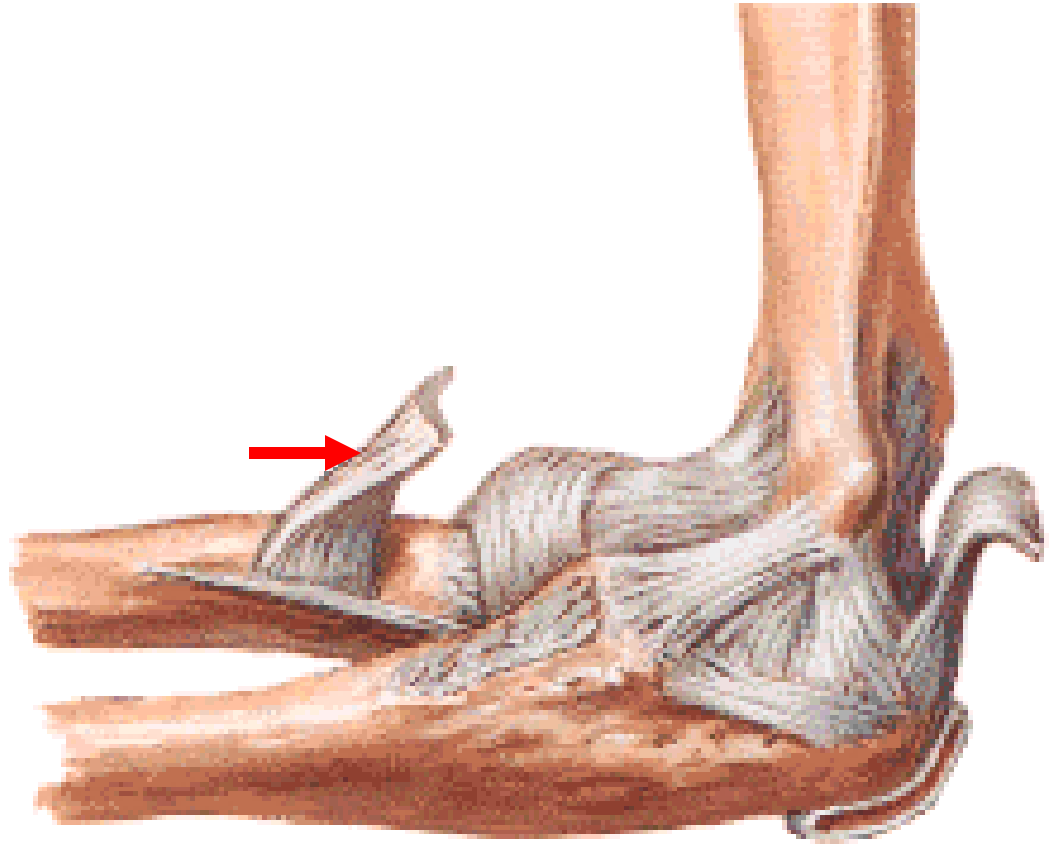
- The muscle lies in front of brachialis with the musculocutaneous nerve sandwiched between them



biceps brachii

□ Insertion

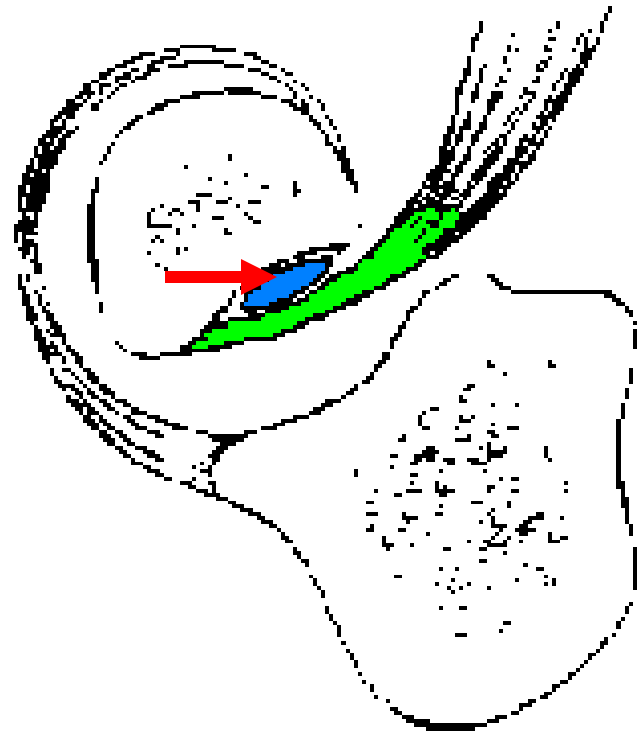
into the radial
tuberosity



Insertion of biceps brachii

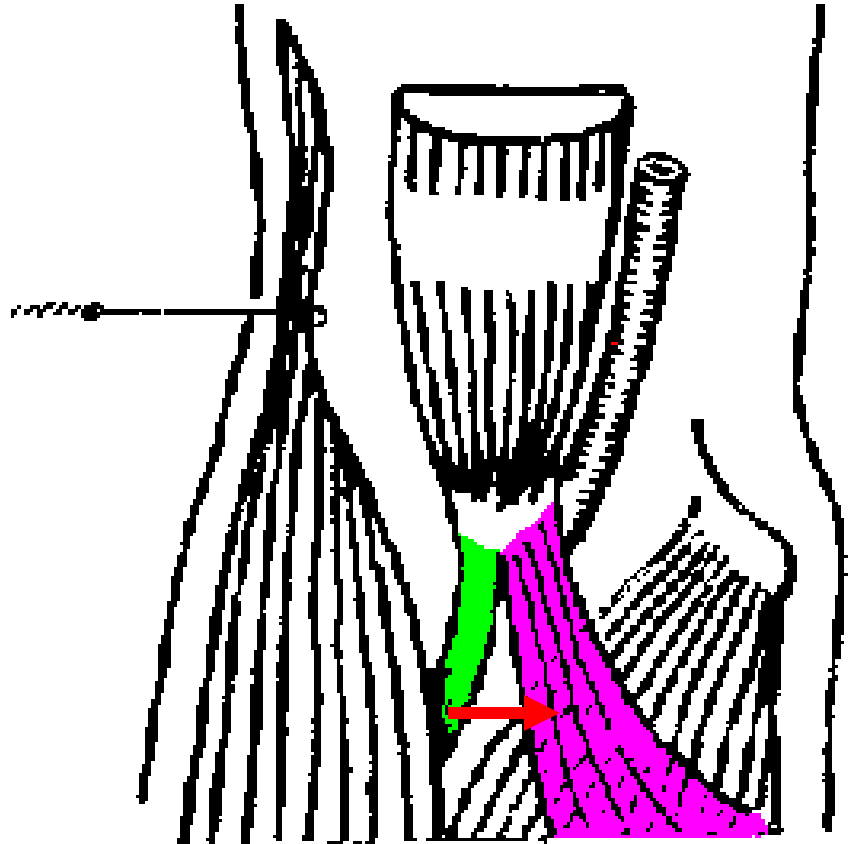
□ A bursa

is present between the tendon and radial tuberosity



Bicipital aponeurosis

- An extension from the tendon of biceps brachii
- passes medially to blend with the deep fascia of the forearm



biceps brachii

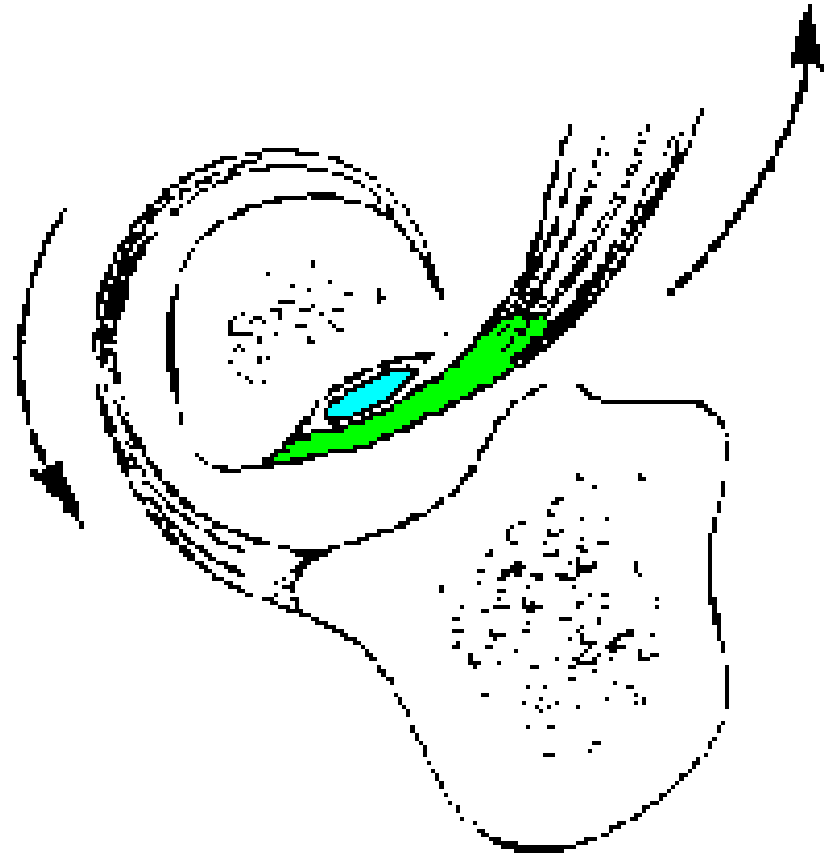
□ Action

Has TWO main actions &
ONE minor action

MAIN

1) because of its insertion
onto the posterior aspect of
the radial tuberosity

It is a powerful supinator of
the forearm

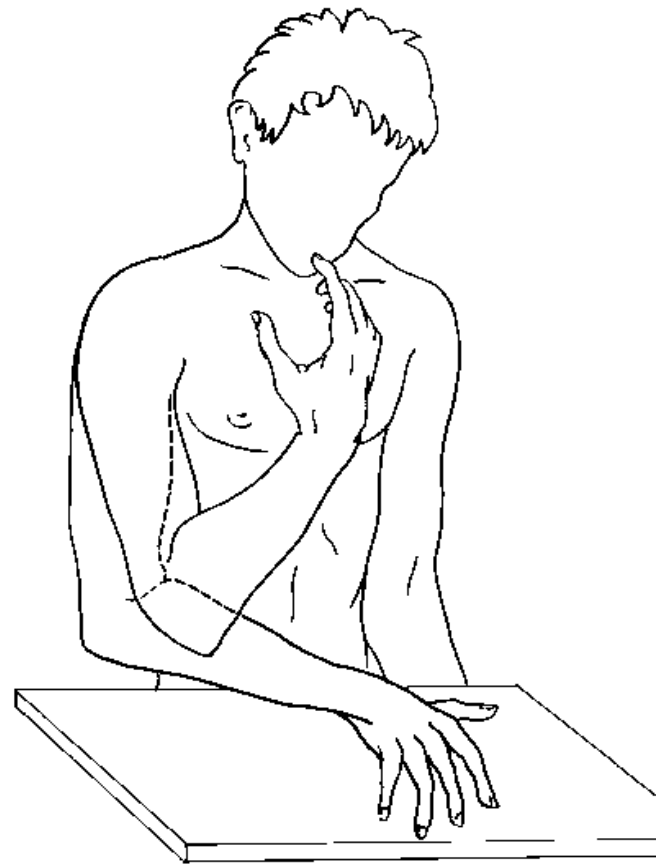


biceps brachii

□ Action

MAIN

2) It is a flexor of the forearm



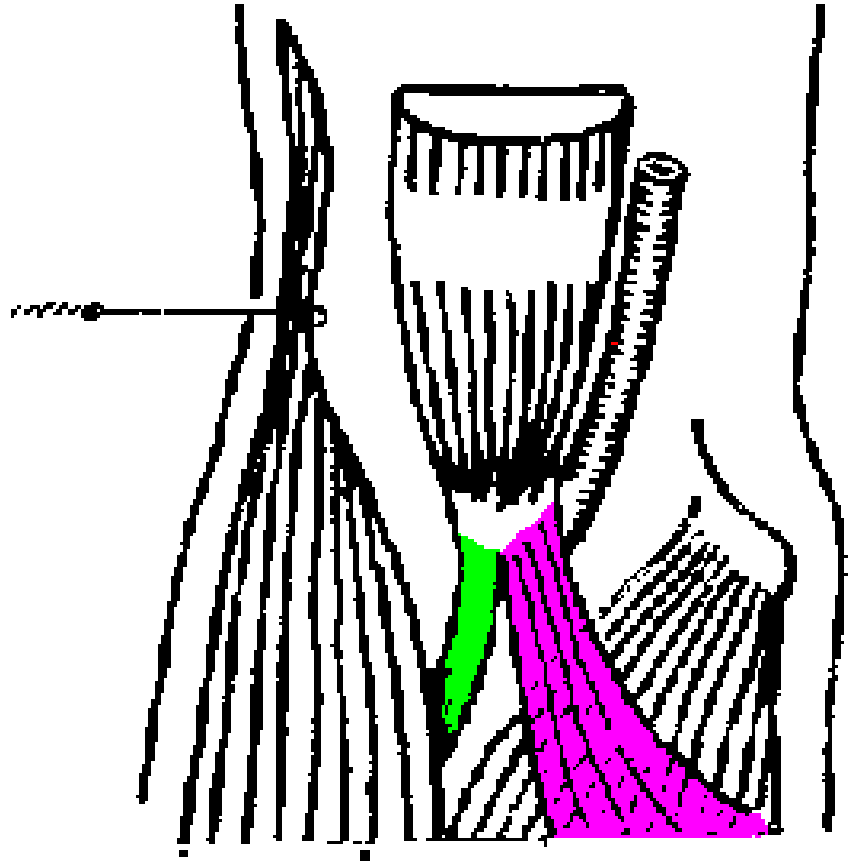
Action of biceps brachii

□ Third action is

its long head arising from the supraglenoid tubercle

a) aids in flexion of the shoulder joint

b) is an important factor in the stability of the abducted shoulder



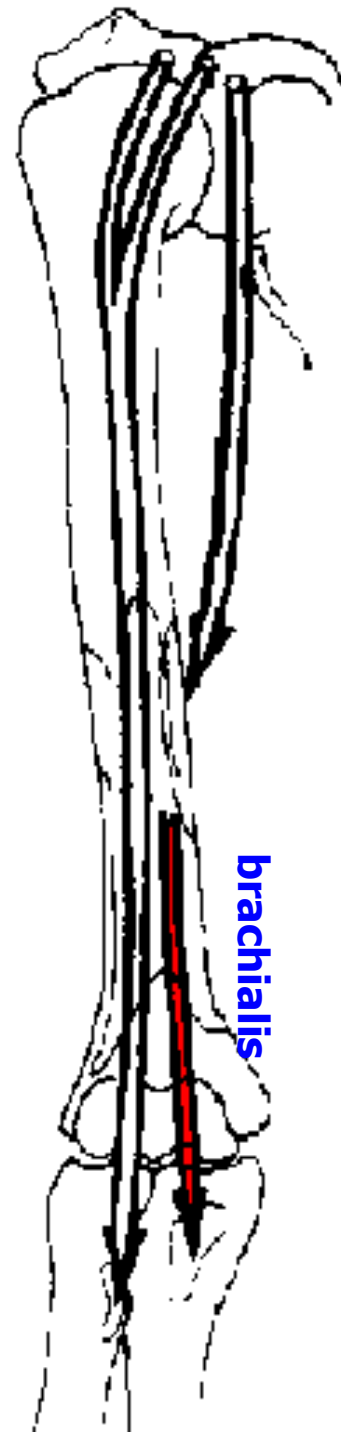
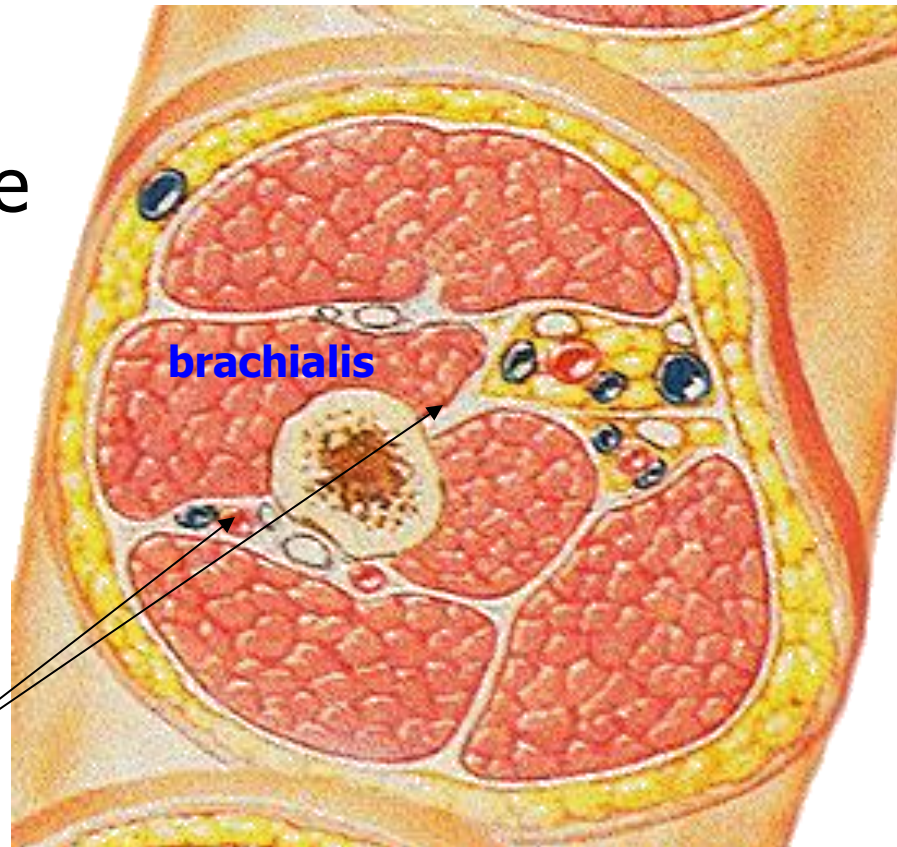
2) Brachialis origin

□ Origin

from:

- front of the lower half of the humerus
- intermuscular septa

Medial & lateral
intermuscular septa

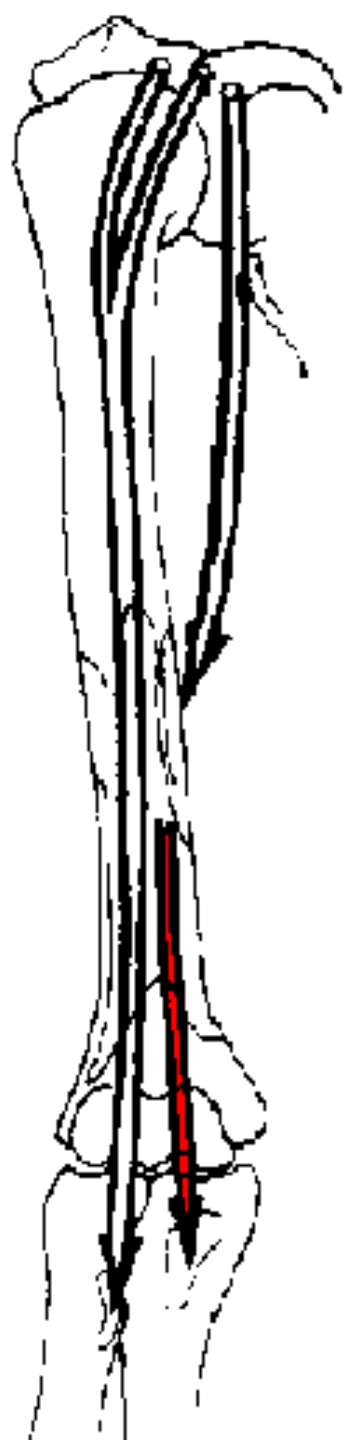


Brachialis insertion

□ Insertion

Into

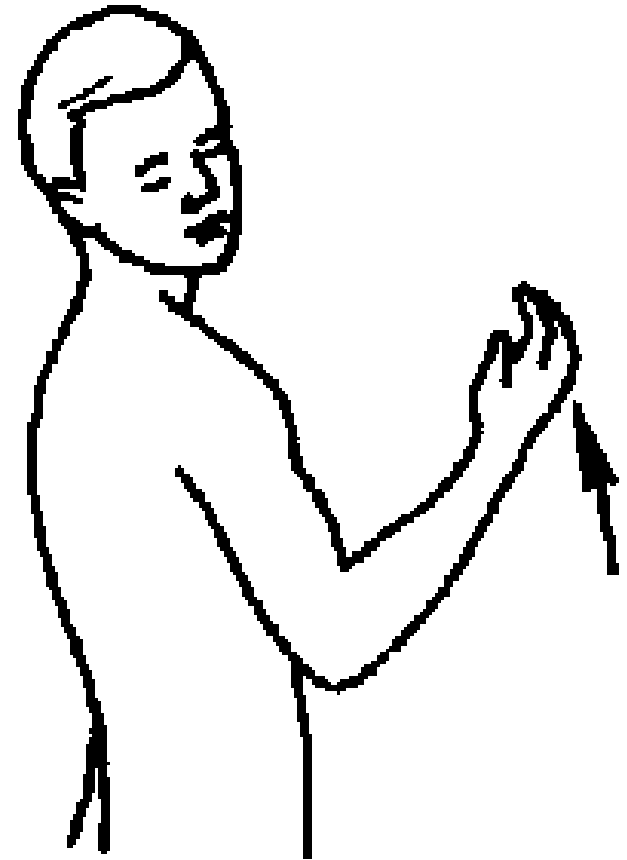
- the coronoid process
- and tuberosity of the ulna.



Brachialis action

□ Action

- Passing in front of the elbow joint
- So it is a powerful flexor of the elbow



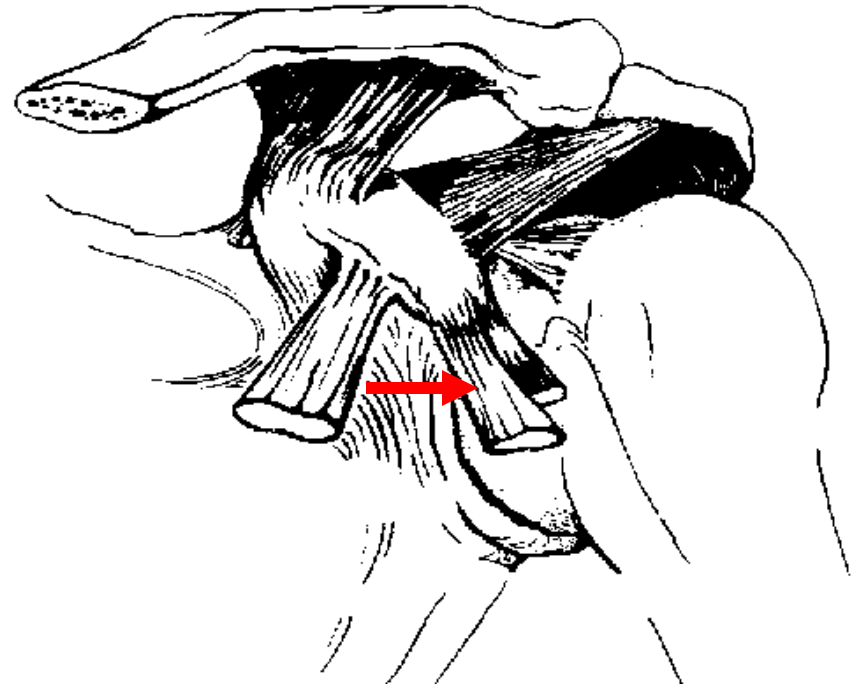
NOTE:

Brachialis has dual innervation (double); Musculocutaneous & radial

3) Coracobrachialis

Origin

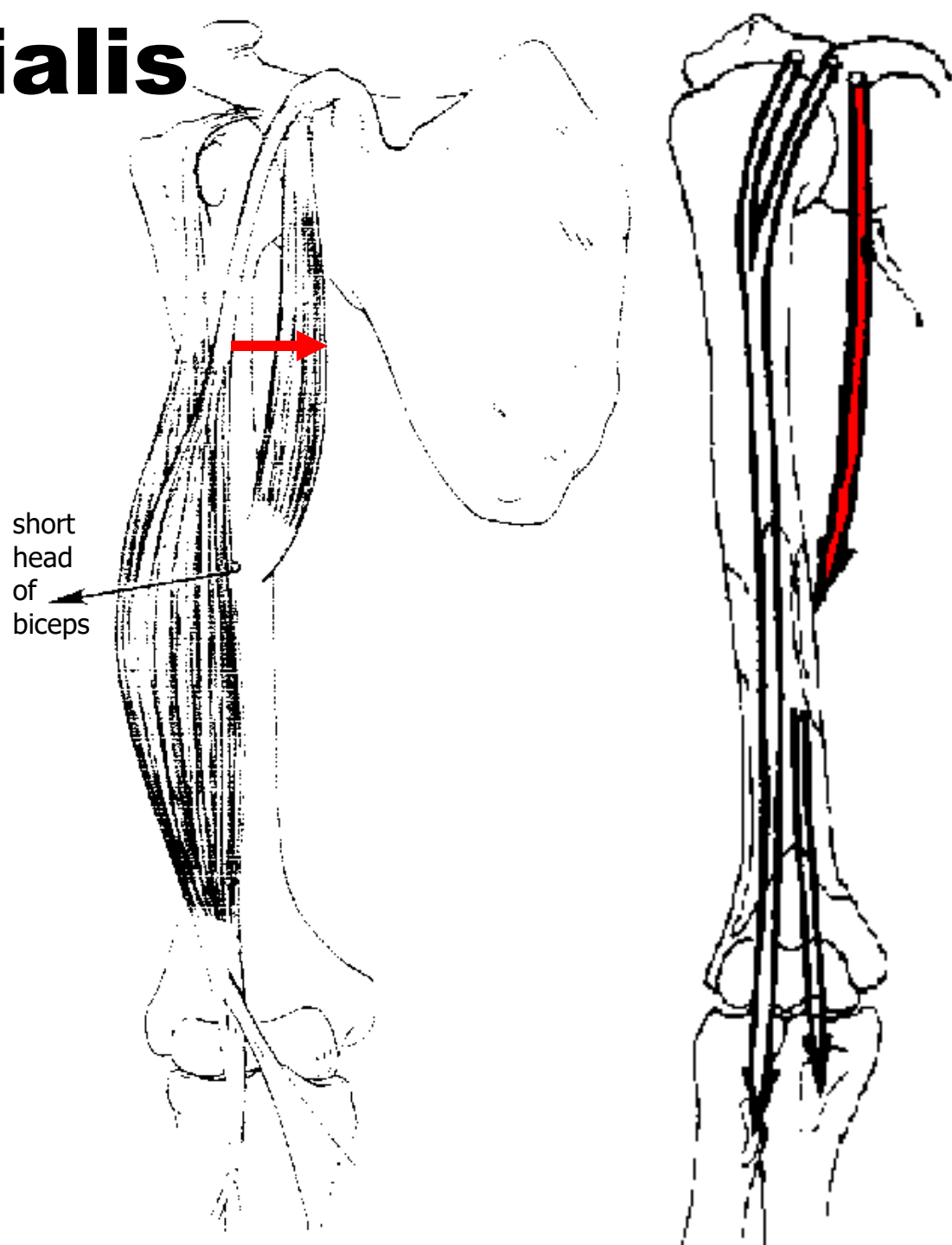
- Is a small muscle
- arises from the tip of the coracoid process in common with the short head of biceps brachii



Coracobrachialis

insertion

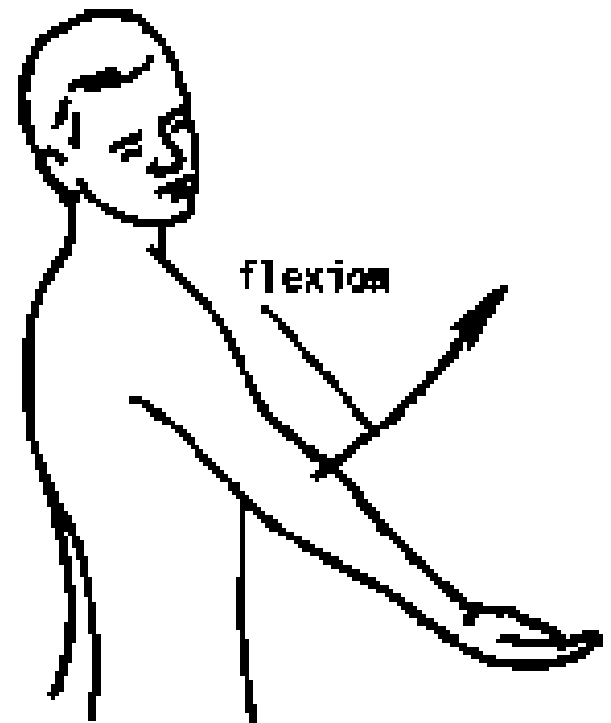
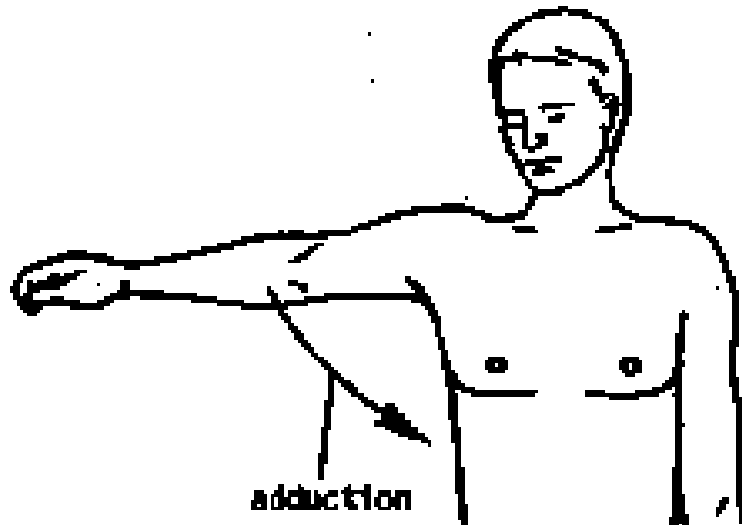
- into the middle point of medial border of the humerus.
- Adjacent to insertion of deltoid at the middle point of lateral border



Coracobrachialis

action

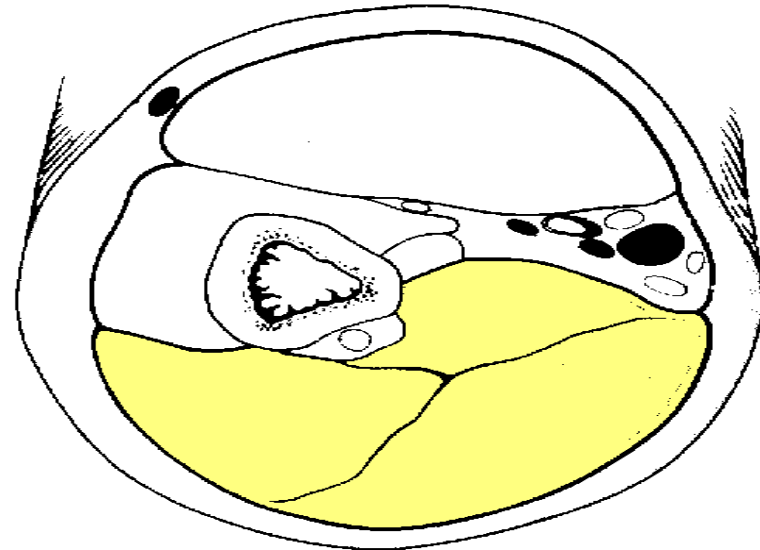
- It is a weak flexor and adductor of the shoulder



POSTERIOR COMPARTMENT

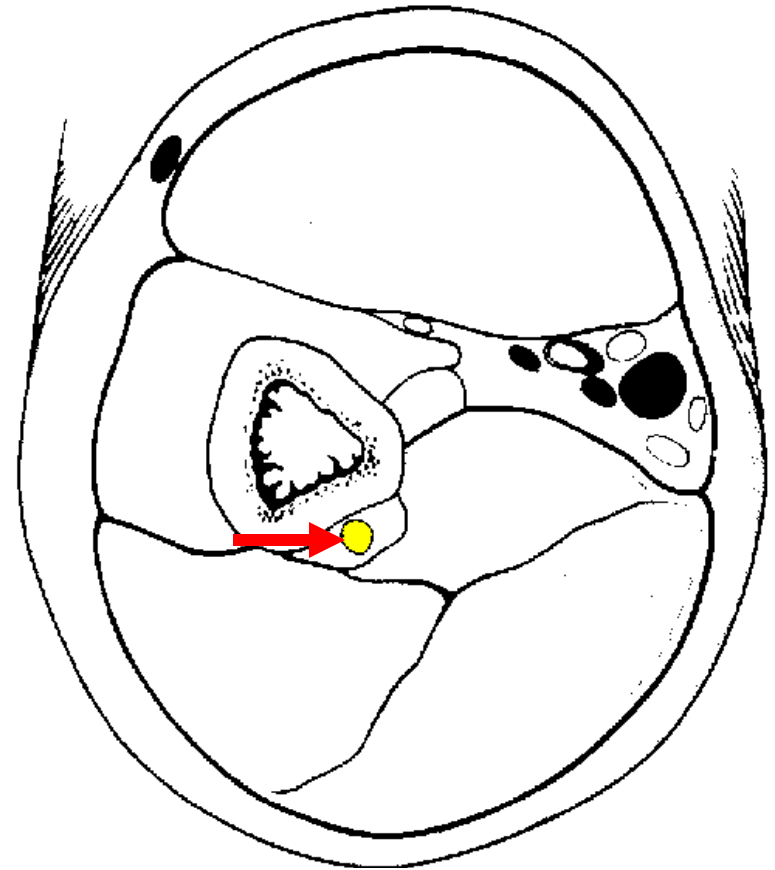
B. Muscles of Back of arm

Muscle	Origin	Insertion	Nerve supply	Action.
Triceps Brachii	a. Long head: infra-glenoid tubercle (extra-capsular) b. Lateral head: Back of shaft of humerus above spiral groove. c. Medial head: Back of shaft of humerus below spiral groove.	Superior surface of olecranon process of ulna.	Radial nerve (long head is supplied in axilla while lateral & medial heads are supplied in spiral groove).	Main extensor of elbow.



Posterior compartment of the arm

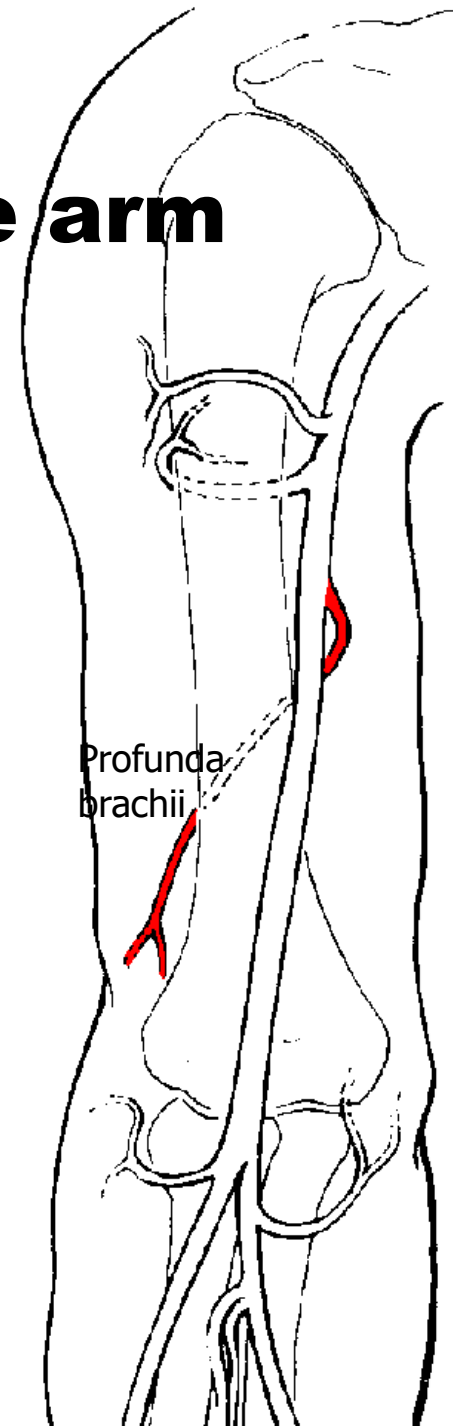
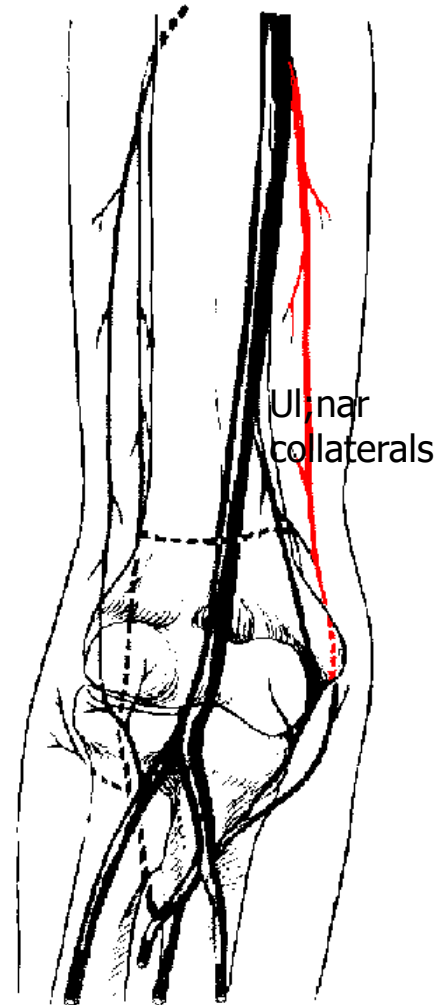
- the nerve of the compartment is the RADIAL NERVE



Posterior compartment of the arm

- The arteries of the compartment are
 - profunda brachii
 - ulnar collateral arteries

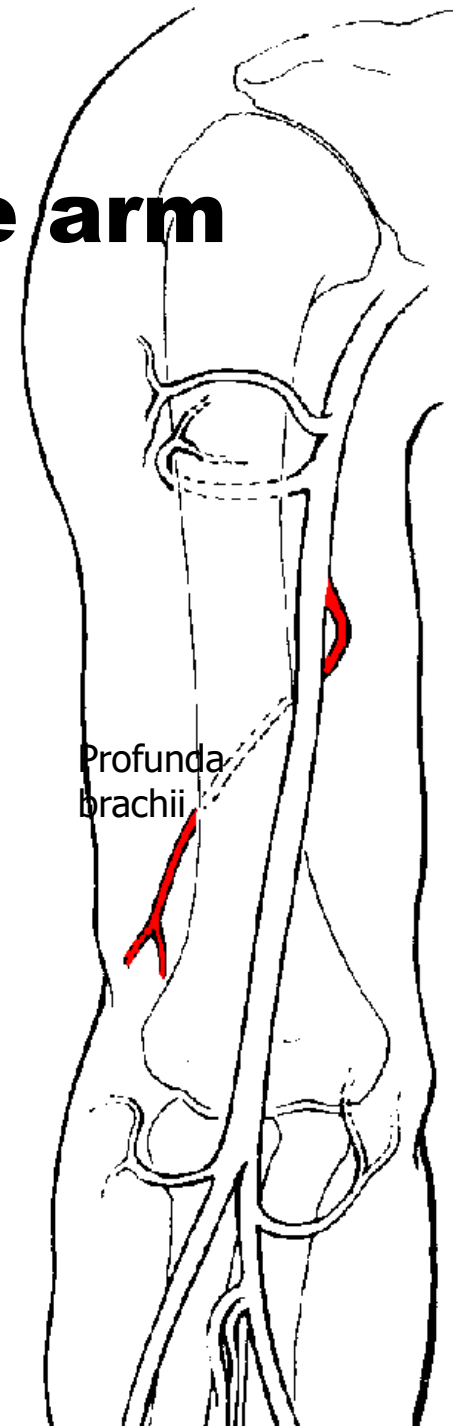
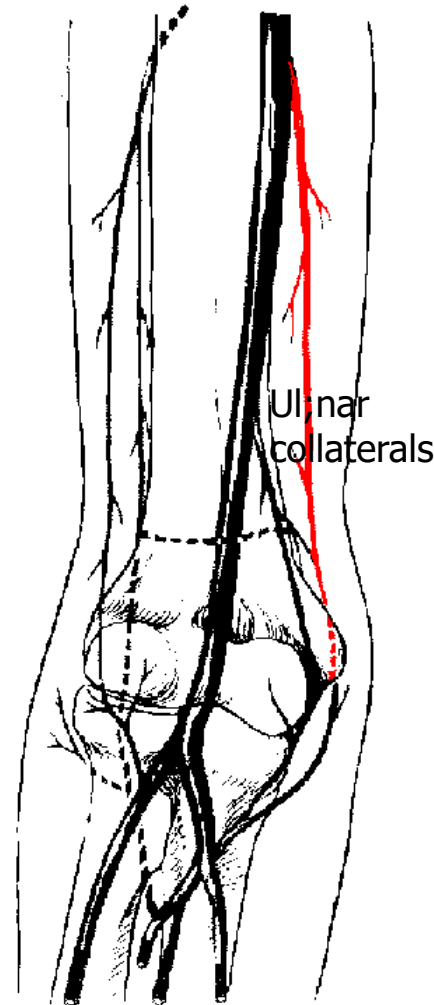
They are branches of the brachial artery



Posterior compartment of the arm

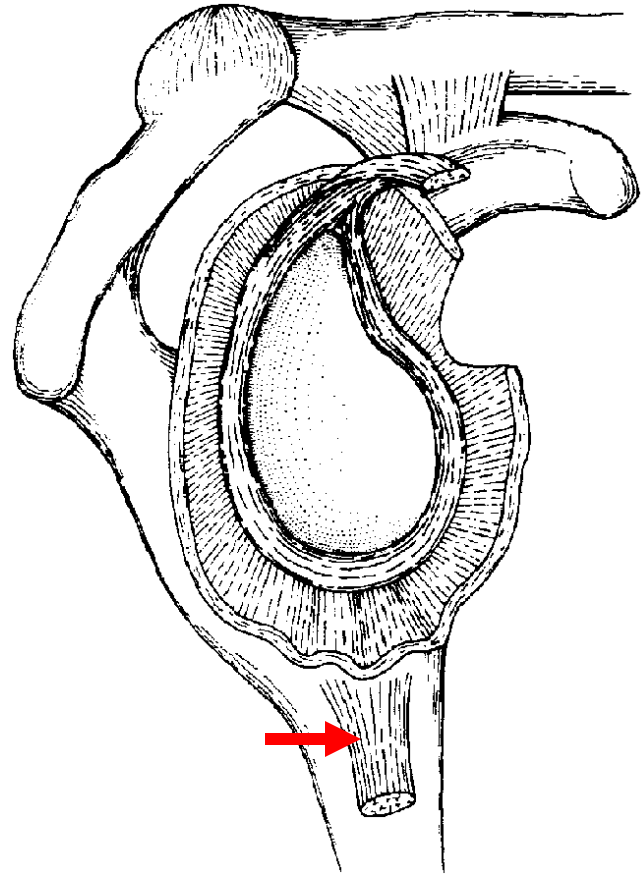
- The arteries of the compartment are
 - profunda brachii
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They are branches of the brachial artery



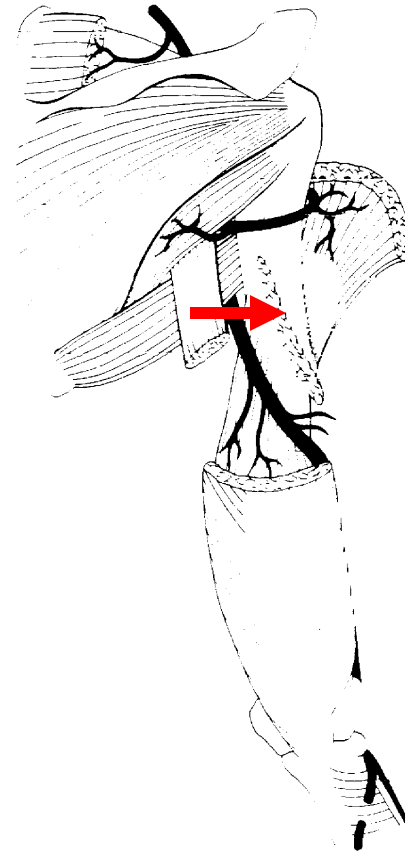
Long head of triceps

- arises from the infraglenoid tubercle of the scapula



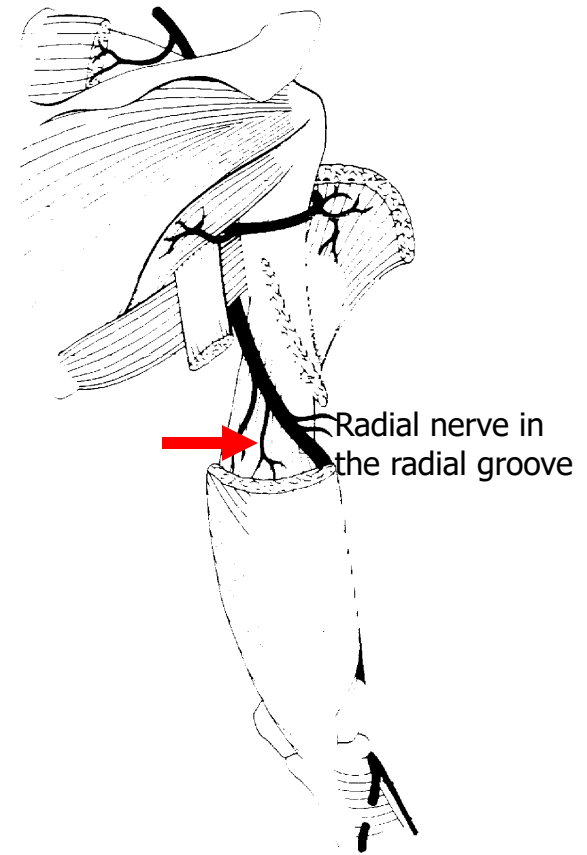
Lateral head of triceps

- arises from the posterior surface of the humerus superior to the radial groove



Medial head of triceps

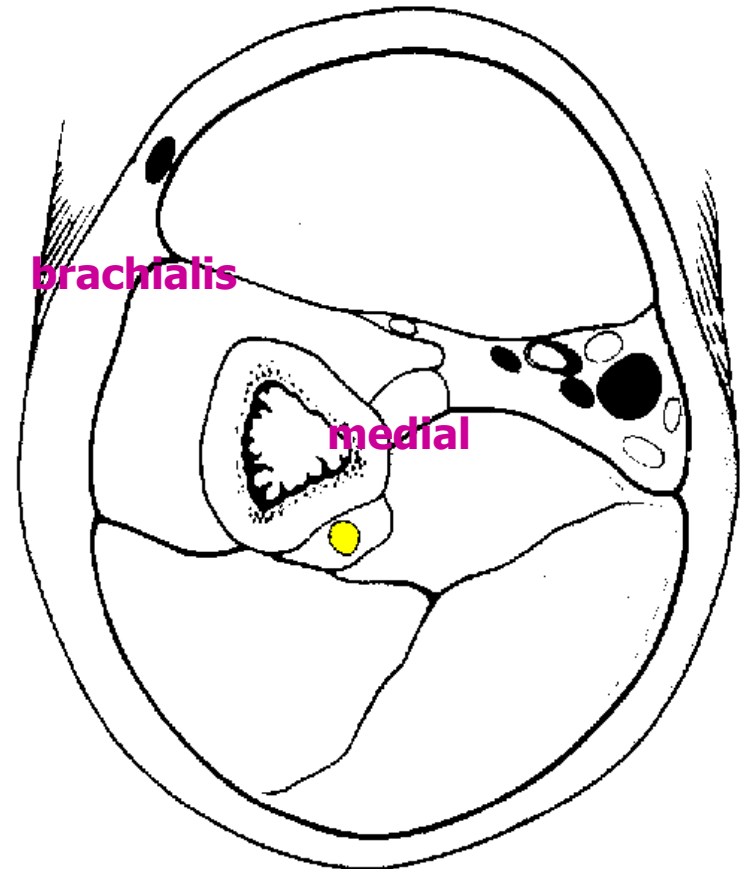
- Arises from the posterior surface of the humerus *inferior* to the radial groove



Triceps heads

□ the medial head is comparable to brachialis

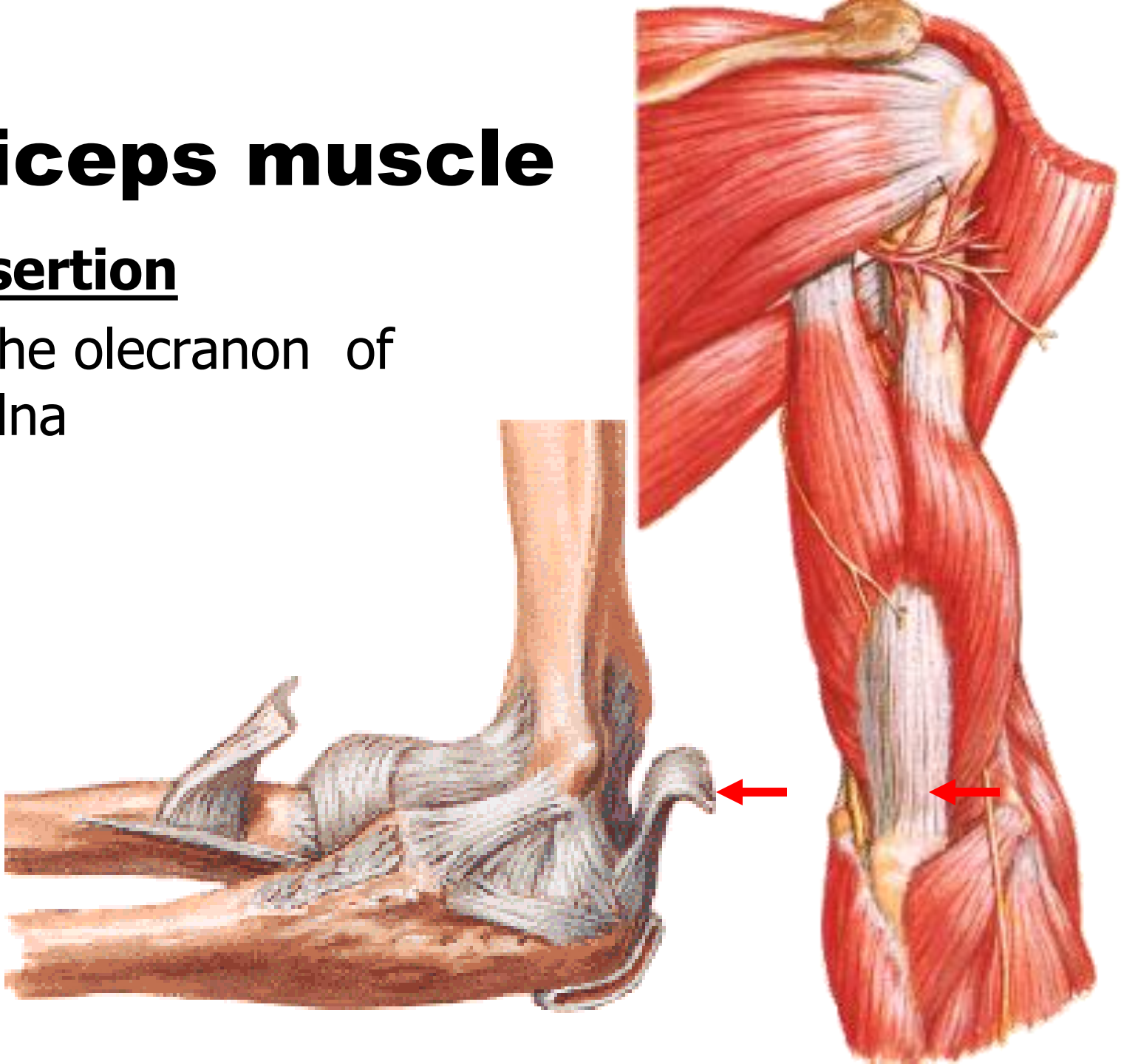
- Lies in a deeper plane than other heads
- takes origin from the lower part of the shaft and the intermuscular septum



Triceps muscle

□ Insertion

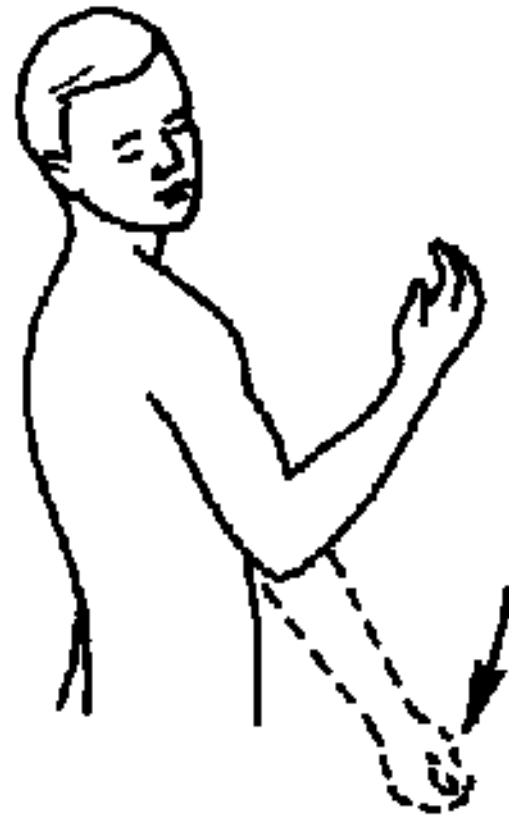
into the olecranon of
the ulna



Triceps action

action

- 1) The muscle is an *extensor* of the elbow



Triceps action

2) its long head arising from the infraglenoid tubercle

a) aids in extending the shoulder joint

b) is an important factor in the stability of the abducted shoulder



Bursae related to triceps tendon

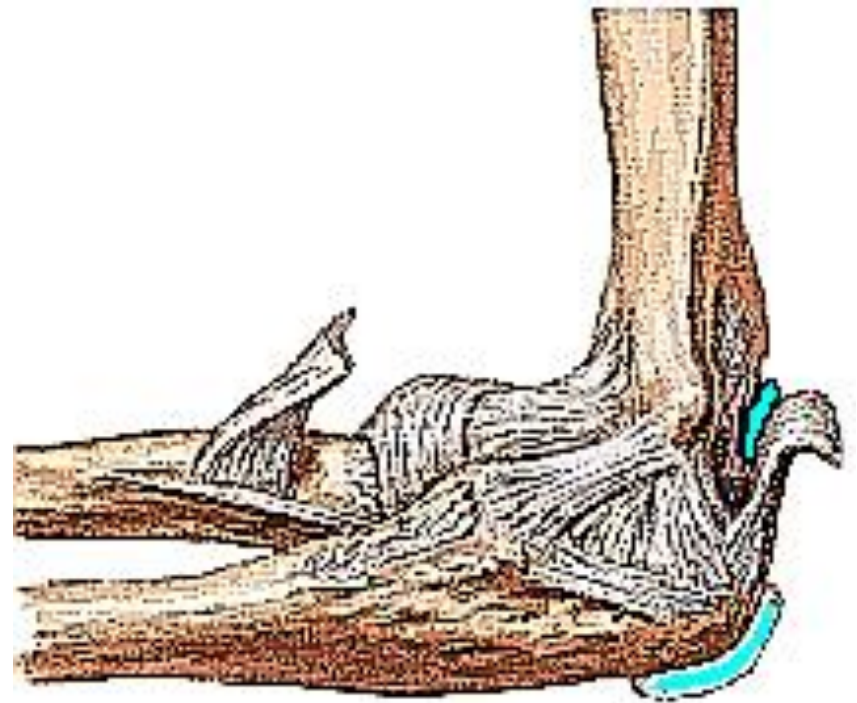
□ It related to TWO bursae

1) Subcutaneous bursa

□ located in the
subcutaneous tissue

2) Subtendinous olecranon
bursa

□ located between the
tendon and olecranon



Student's elbow

- The subcutaneous bursa is liable to be inflamed by repeated friction this is called "student's elbow "



ELBOW JOINT

ELBOW JOINT

❖ Type & Variety ;

- synovial of Hinge variety (uniaxial)
- Compound joint (3 bones)

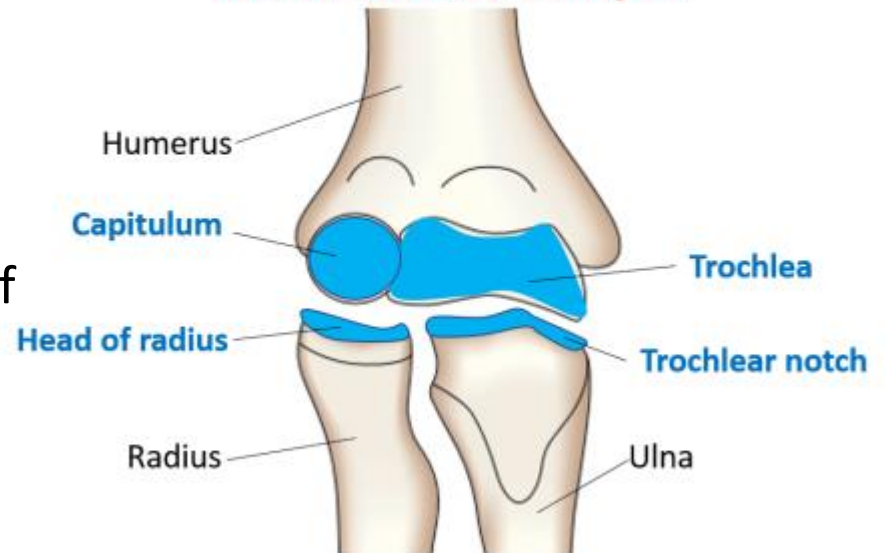
❖ Articular surfaces:

- Trochlea & capitulum of the lower end of the humerus
- Trochlear notch of ulna & head of radius

❖ Capsule:

- thin in front & behind
- thick at side

Articular surfaces of Elbow joint



Attachment of Capsule of Elbow joint



ELBOW JOINT

❖Ligament :

- as every hinge joint supported by 2 strong collateral ligament

1) radial collateral ligament

- ✓ Strong triangular ligament on the late side
- ✓ Apex attach to lat. Epicondyle
- ✓ Base to annular ligament

2) Ulnar collateral ligament

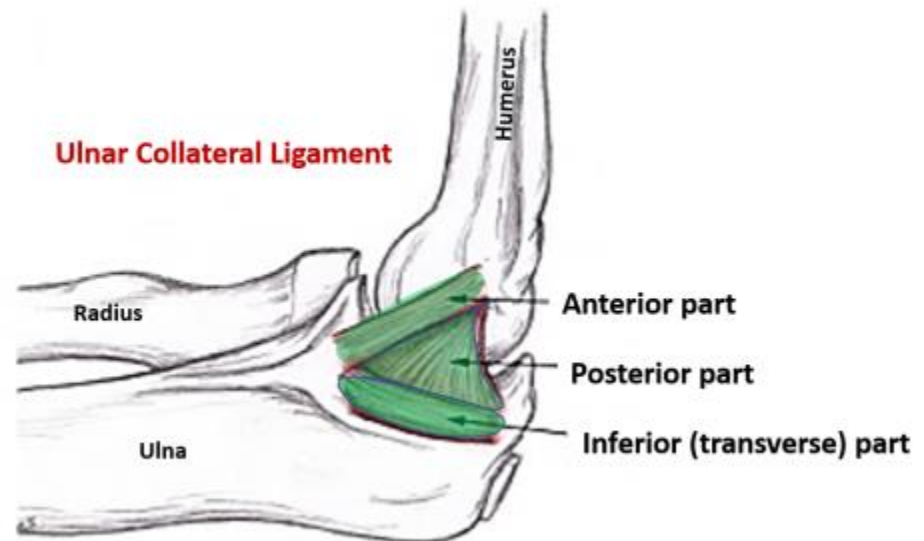
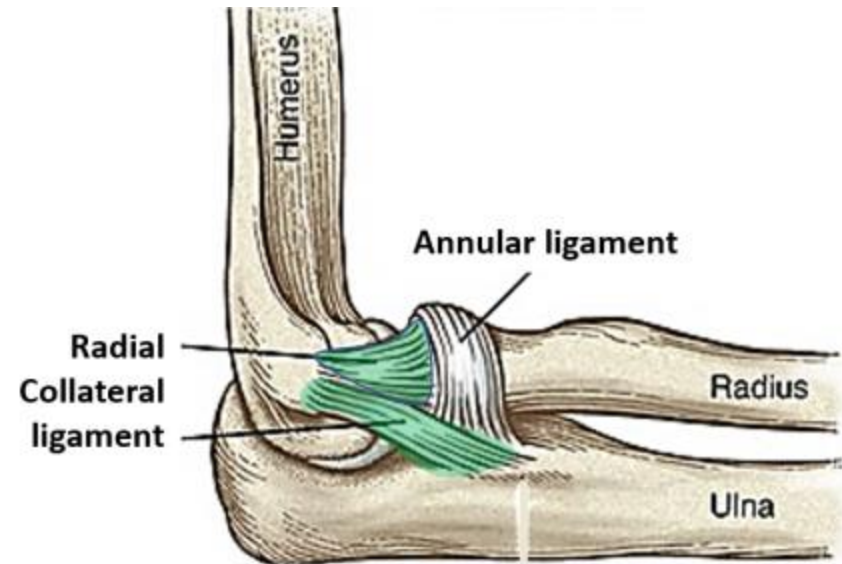
- ✓ Strong triangular ligament on the med. Side
- ✓ Apex attach to med.. Epicondyle
- ✓ Base extends from coronoid process to olecranon process

N.B :

Ulnar collateral ligament is **crossed by ulnar n.**

❖Movement;

Flexion ,extension



SUPERIOR(PROXIMAL) RADIOULNAR JOINT

proximal radio-ulnar joint

❖Type & Varity :

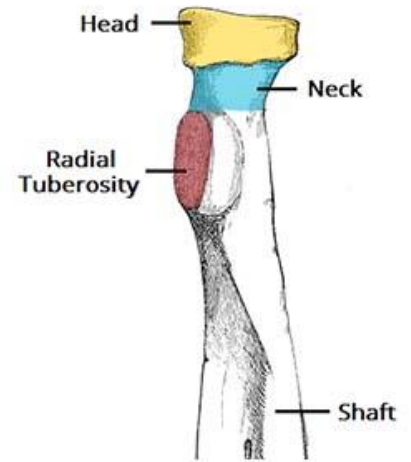
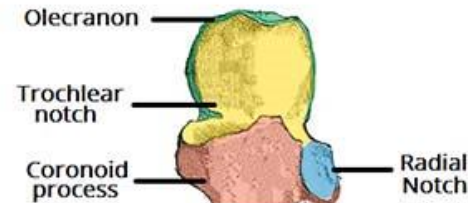
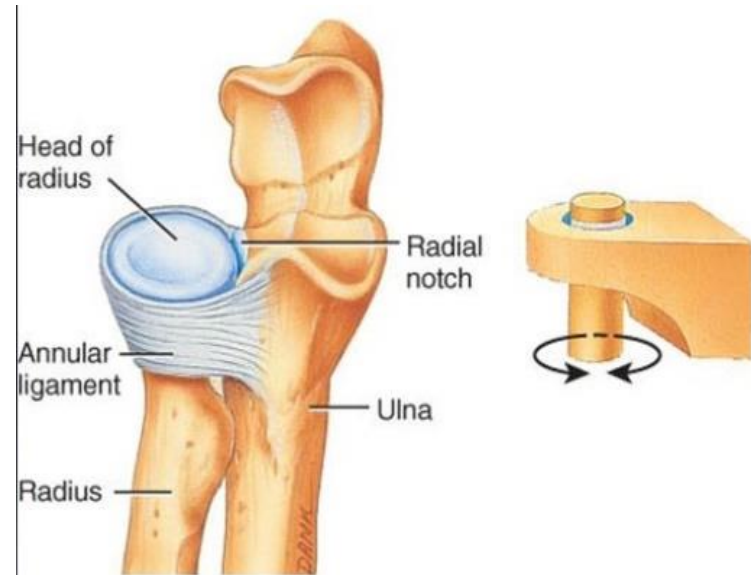
Synovial, uniaxial (pivot) joint.

Articular surfaces

- 1- head of radius
- 2- radial notch of ulnar & annular ligament

❖Capsule;

- Covers the annular ligament
- Above Continuous with capsule of elbow joint. **Their joint spaces are continuous together**
- Below loosely attached to the neck of the radius
- The synovial membrane is a downward prolongation of the synovial membrane of the elbow joint



proximal radio-ulnar joint

❖ Ligaments :

1) annular ligament

- strong curved
- form $\frac{3}{4}$ of the circle With radial notch

2) Quadrate ligament

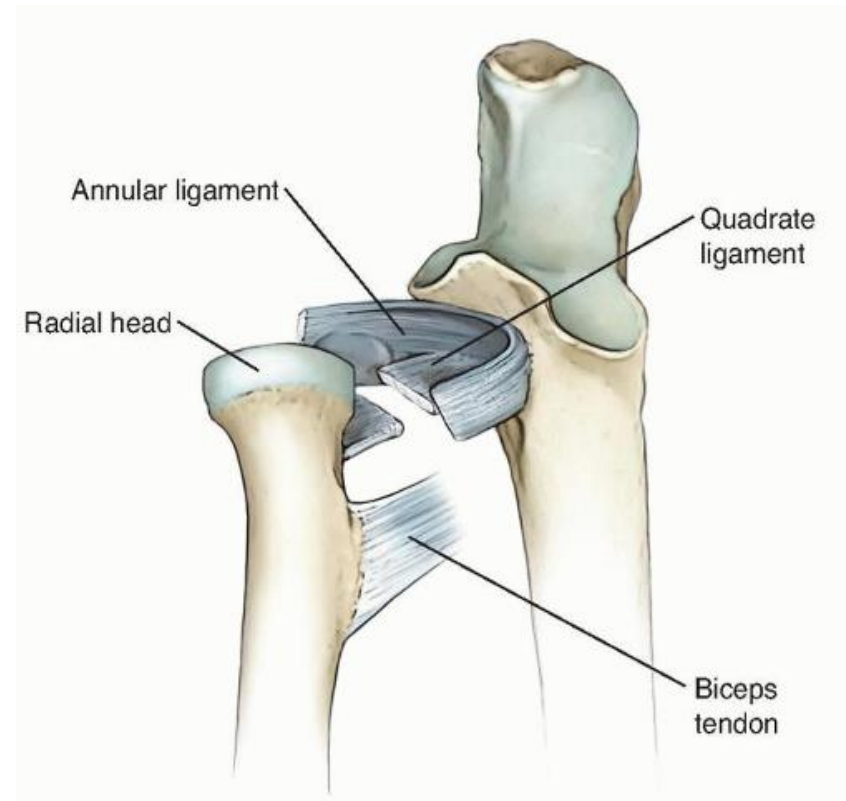
- thin quadrangular fibrous band
- extends from lower border of radial notch of ulna medially to the neck of radius laterally
- closing the joint infromedially

N.B:

cavity of proximal radioulnar joint is continuous with elbow joint cavity above

❖ Movement

- supination & pronation



1) It is initiated by brachioradialis (rotates the radius from full **SUPINATION** to midprone position).

2) It is completed by

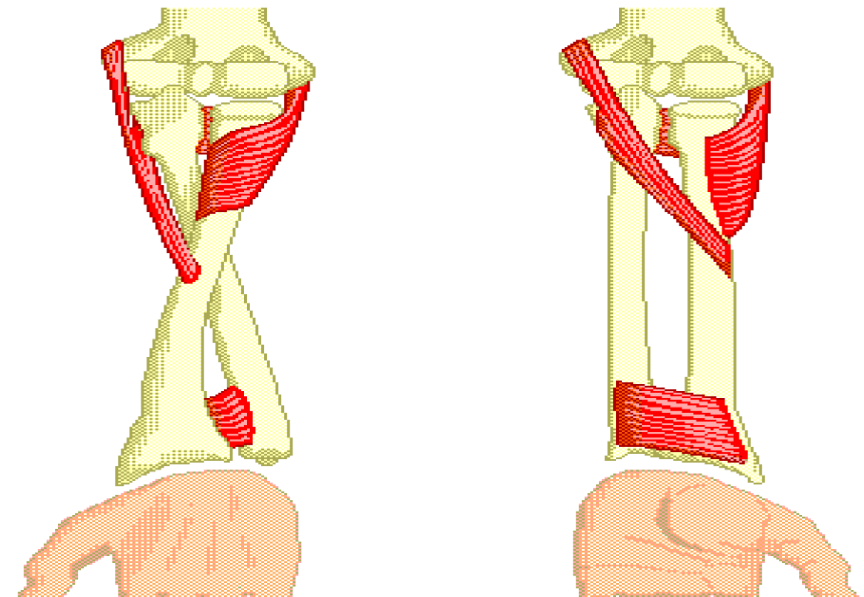
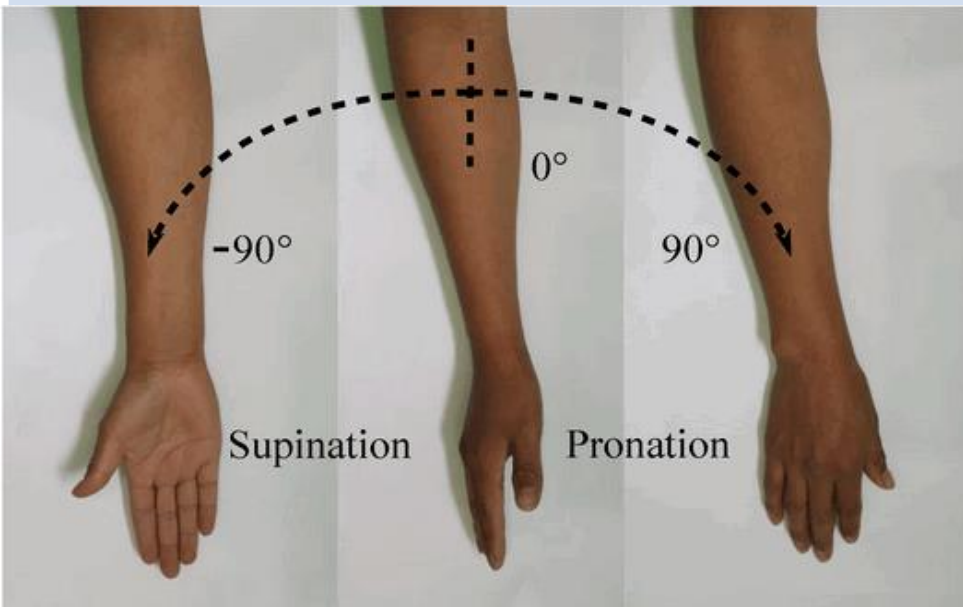
- Pronator teres
- pronator quadratus

1) It is initiated by brachioradialis (rotates the radius from full **PRONATION** to midprone position).

2) It is completed by

- supinator (when elbow is extended)
- Biceps (when elbow is flexed)

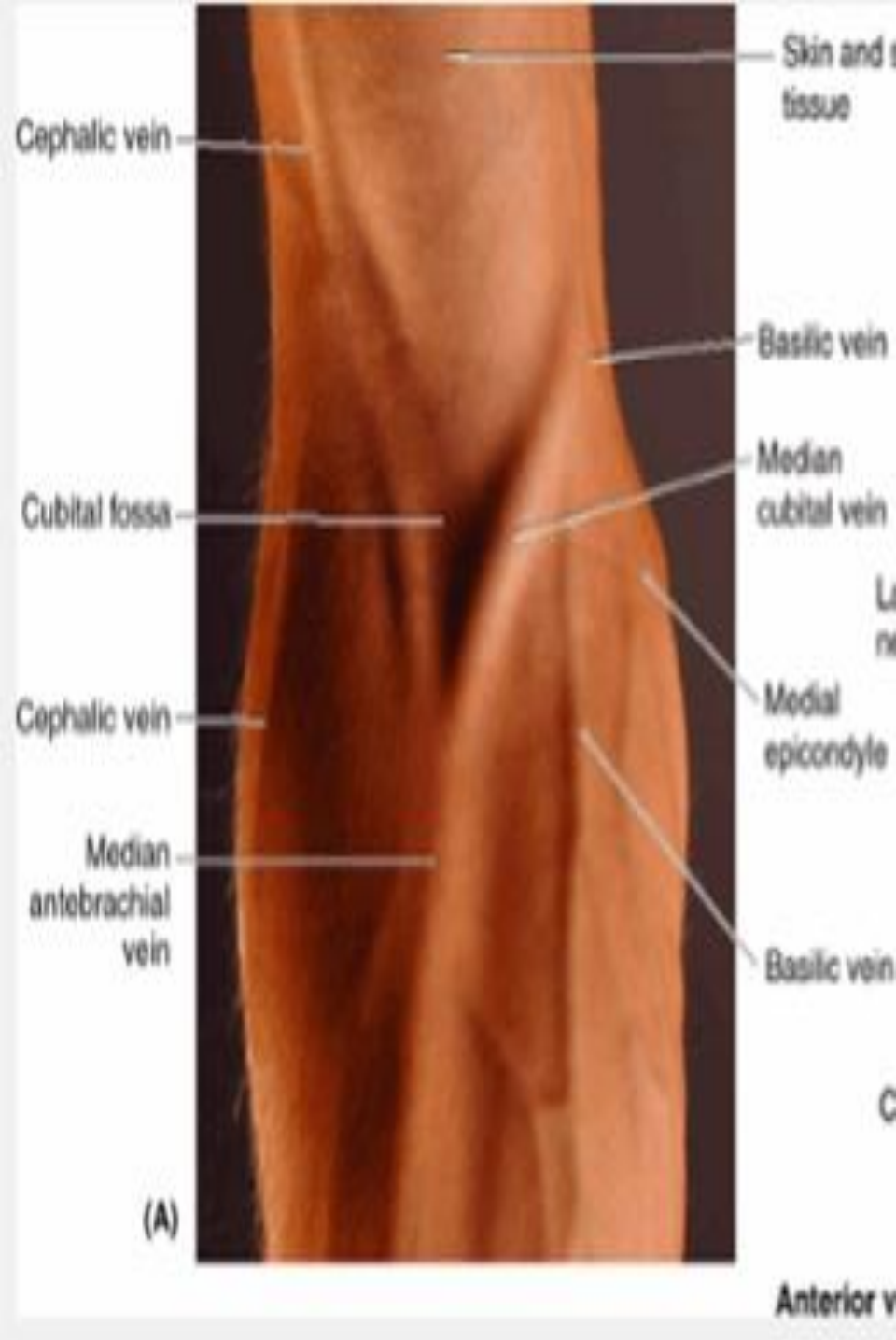
Axis of movement is ULNA



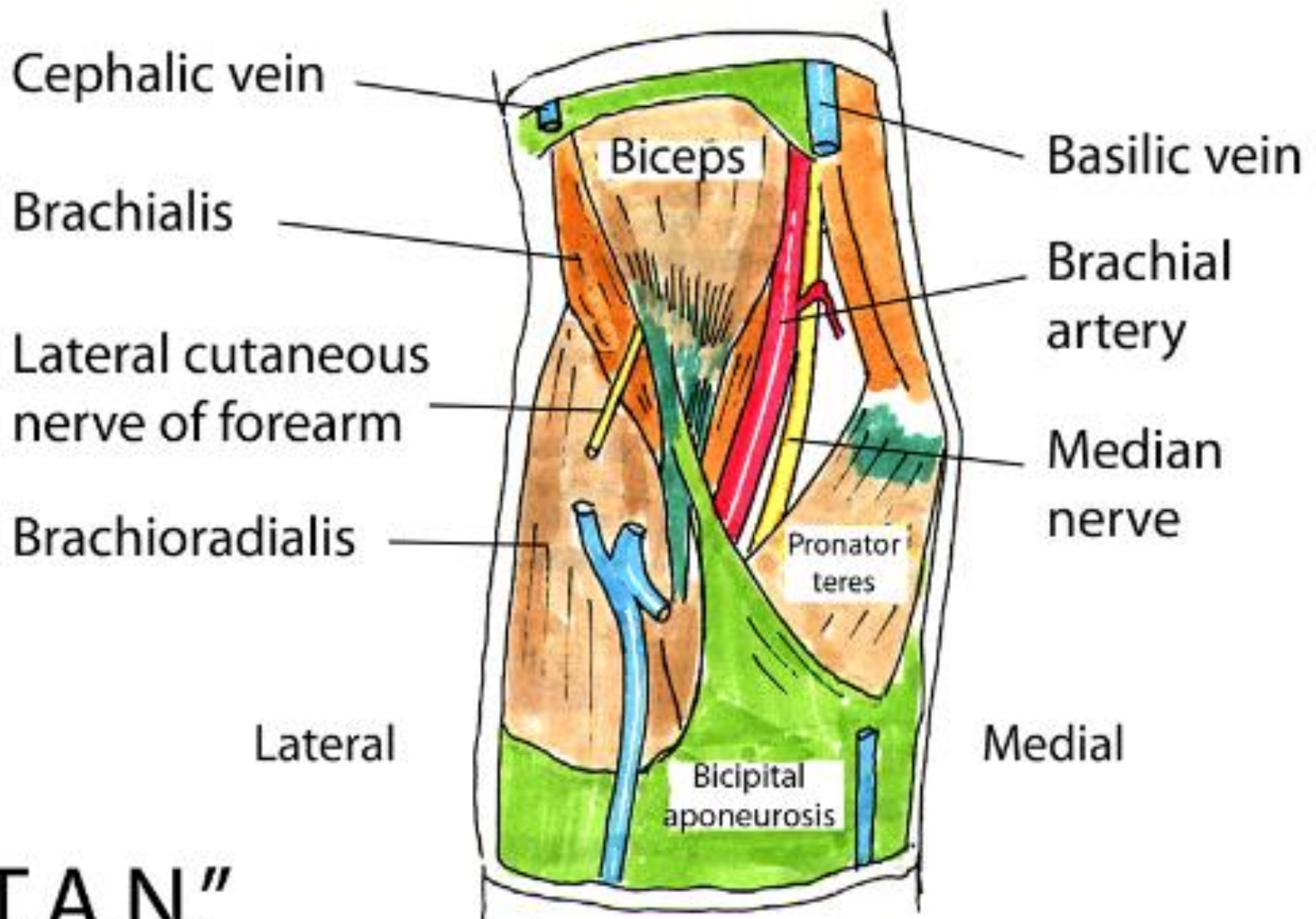
CUBITAL FOSSA

CUBITAL FOSSA

- Site
- Shape
- Boundaries
 - Medial Border
 - Lateral Border
 - Apex
 - Base
 - Floor
 - Roof
- Contents

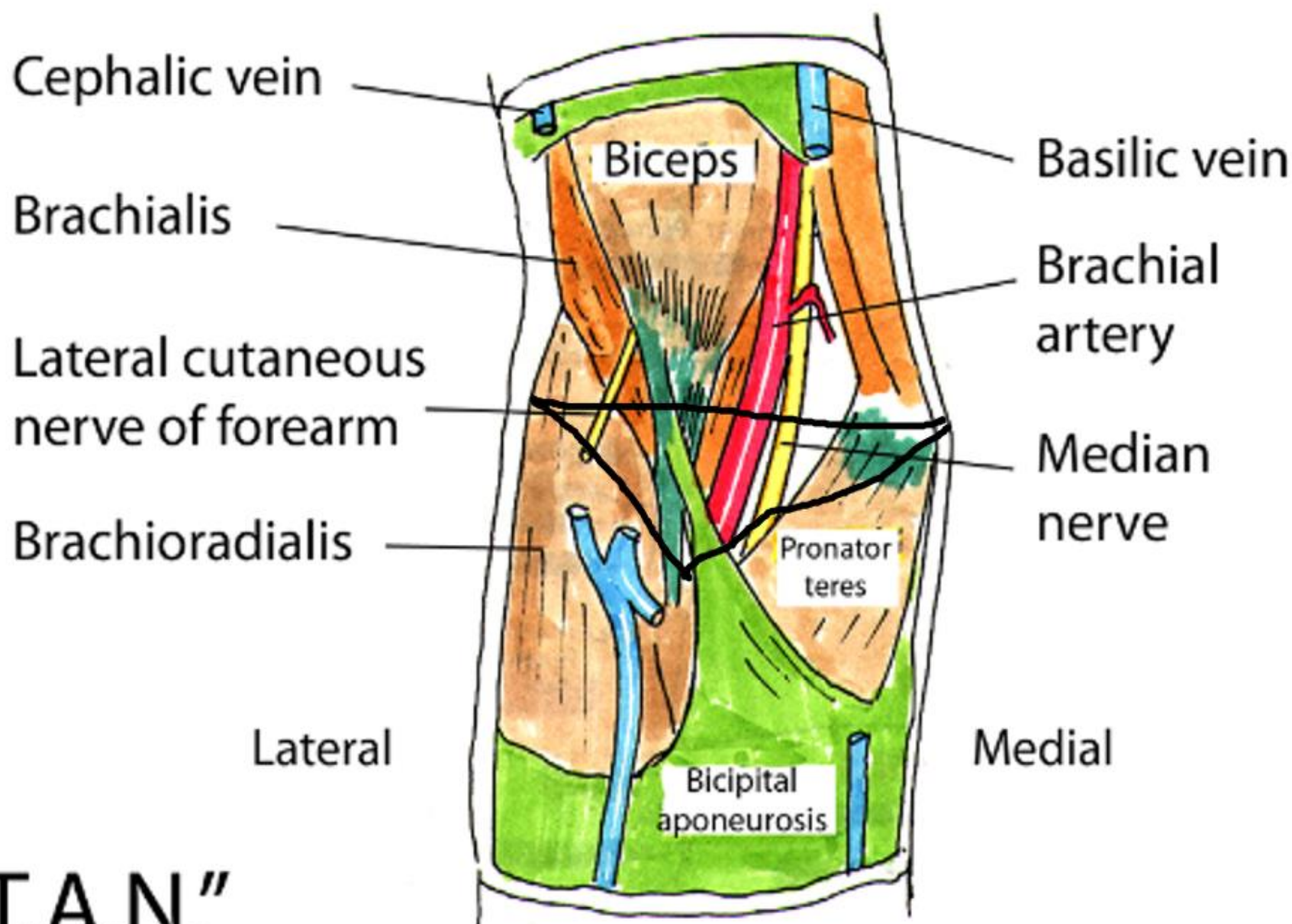


CUBITAL FOSSA



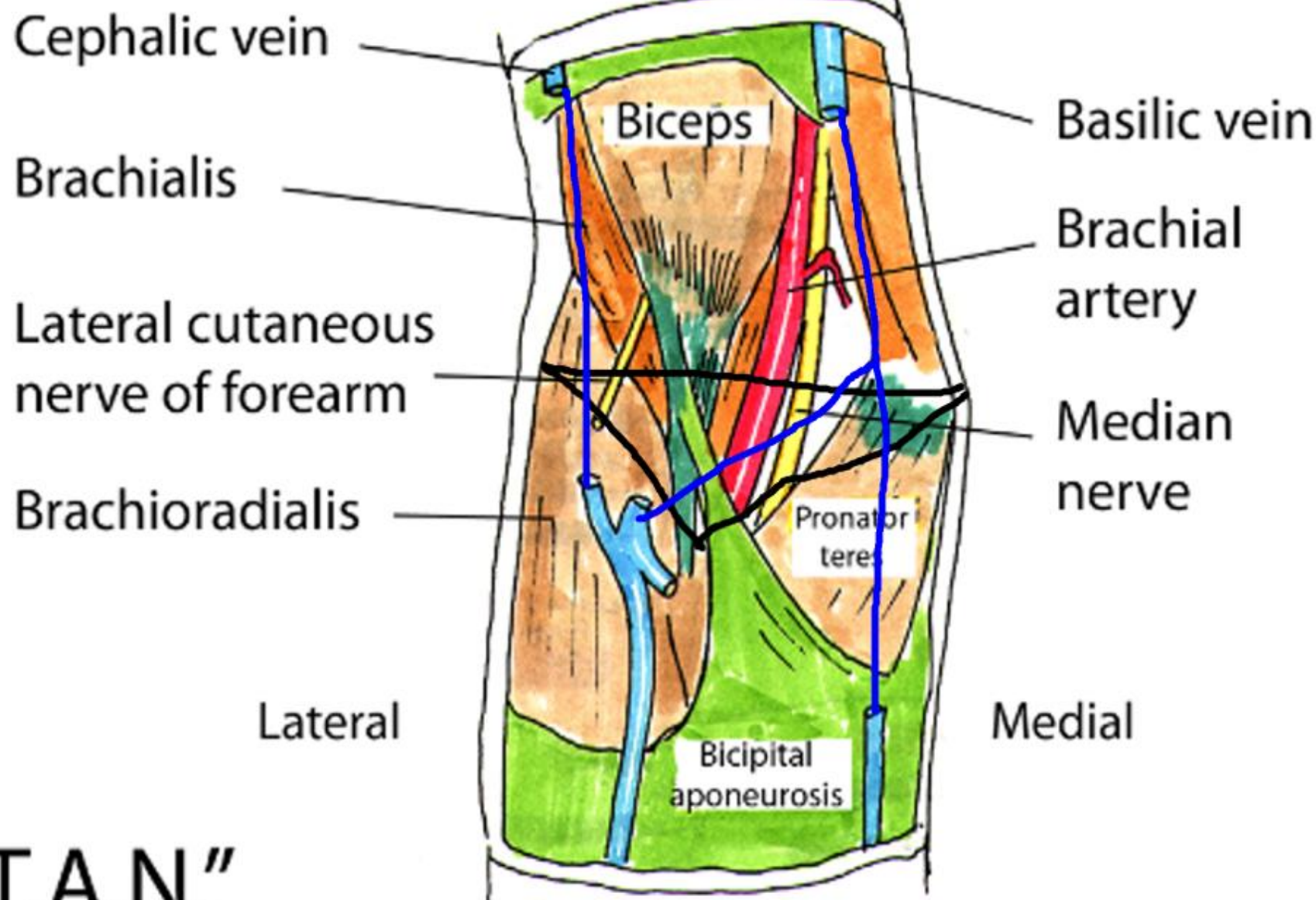
"T.A.N."

CUBITAL FOSSA

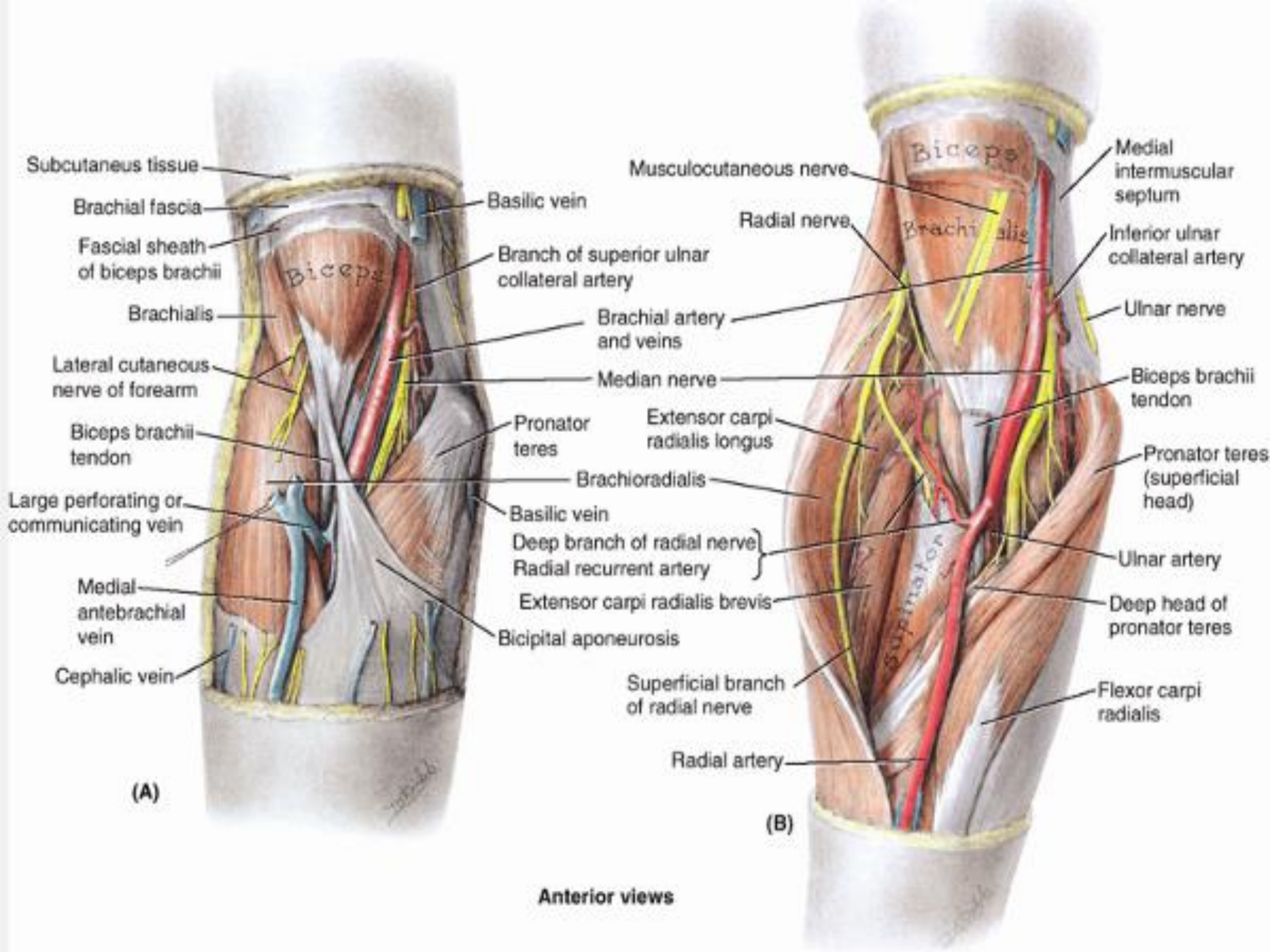


"T.A.N."

CUBITAL FOSSA



"T.A.N."



CUBITAL FOSSA

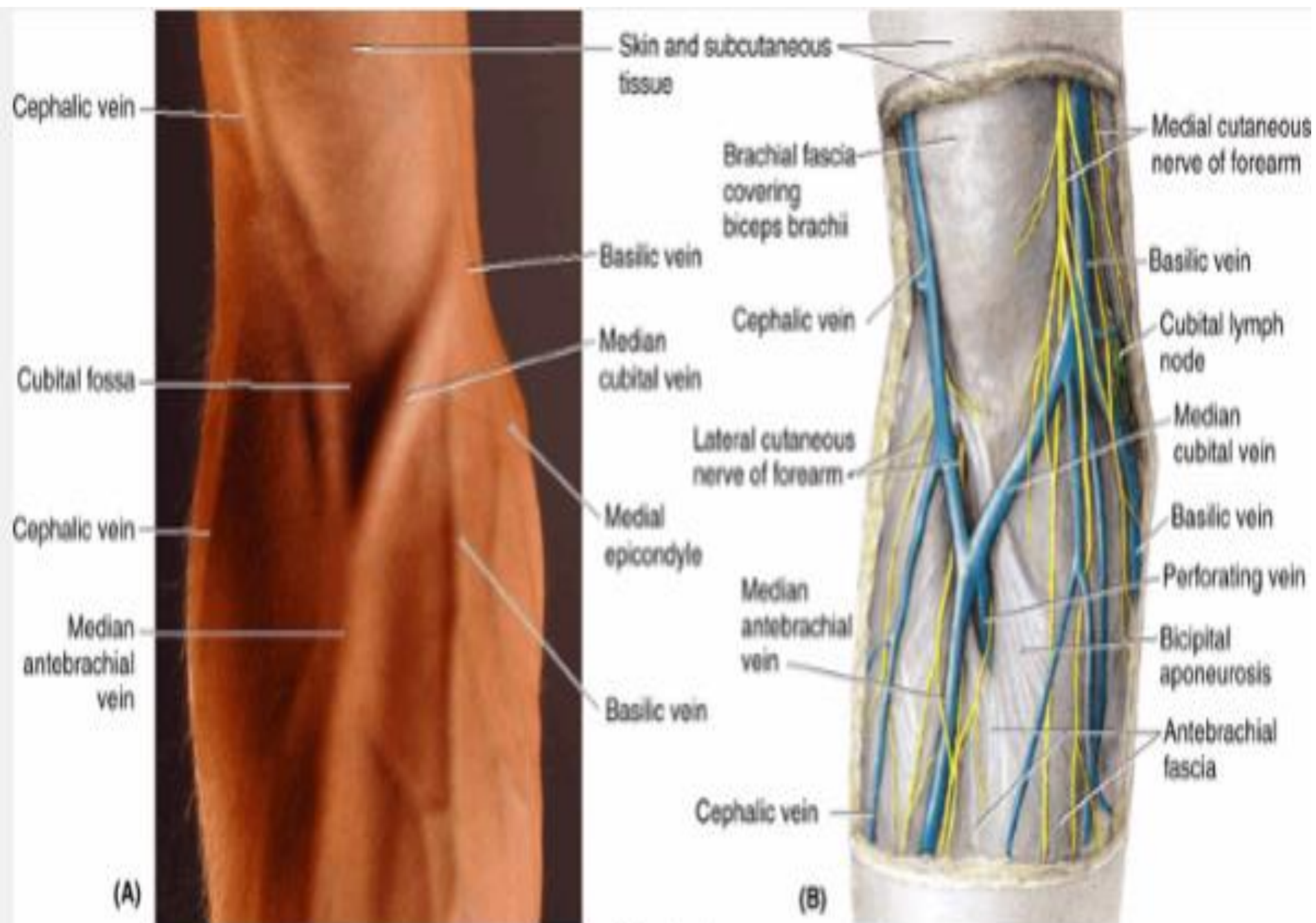
- **Definition & Site:** intermuscular space in front of elbow
- **Shape:** triangular in shape, its base above & apex below
- **Boundaries:**
 - ✓ **Base:** intercondylar line between to TWO humeral epicondyles
 - ✓ **Medial border** : pronator teres m.
 - ✓ **Lateral border:** brachioradialis m
 - ✓ **Apex:** formed by the meeting of the medial & lateral borders
 - ✓ **floor** : formed by
 - 1- brachialis (above & medially)
 - 2- supinator (below & laterally)
 - ✓ **Roof:** formed by
 - 1-skin
 - 2- superficial fascia containing
 - a) basilic v. with med. Cut. N. of forearm (medially)
 - b) cephalic v. lat. Cut. N. of forearm (laterally)
 - c) median cubital v. connecting the basilic & cephalic veins
 - 3- deep fascia (reinforced by the bicipital aponeurosis)

CUBITAL FOSSA

□ **Contents:** arranged from medial to lateral as follows:

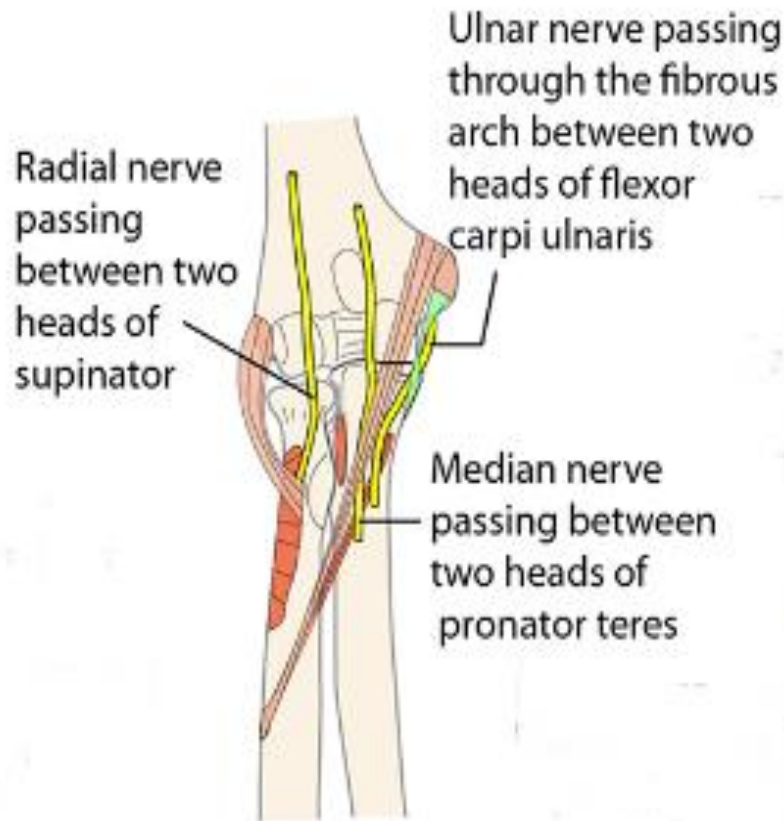
- 1- **median n.:** leaves the fossa by passing between the 2 heads of pronator teres
- 2- **the terminal part of brachial a.** & the beginning of both radial & ulnar arteries
 - a) radial a. leaves the fossa by passing superficial to pronator teres
 - b) ulnar a. leaves the fossa by passing deep to pronator teres m.
- 3- **biceps tendon** passes backwards to reach its insertion in the radial tuberosity
- 4- **radial n. & its post. Interosseous br.**
 - **Radial n.** leaves the fossa by passing deep to the brachioradialis m.
 - **Post. Interosseous n.** leaves the fossa by piercing the supinator m.

N.B: cubital fossa contains one LN (epitrochlear LN) lying in front of the med. epicondyle



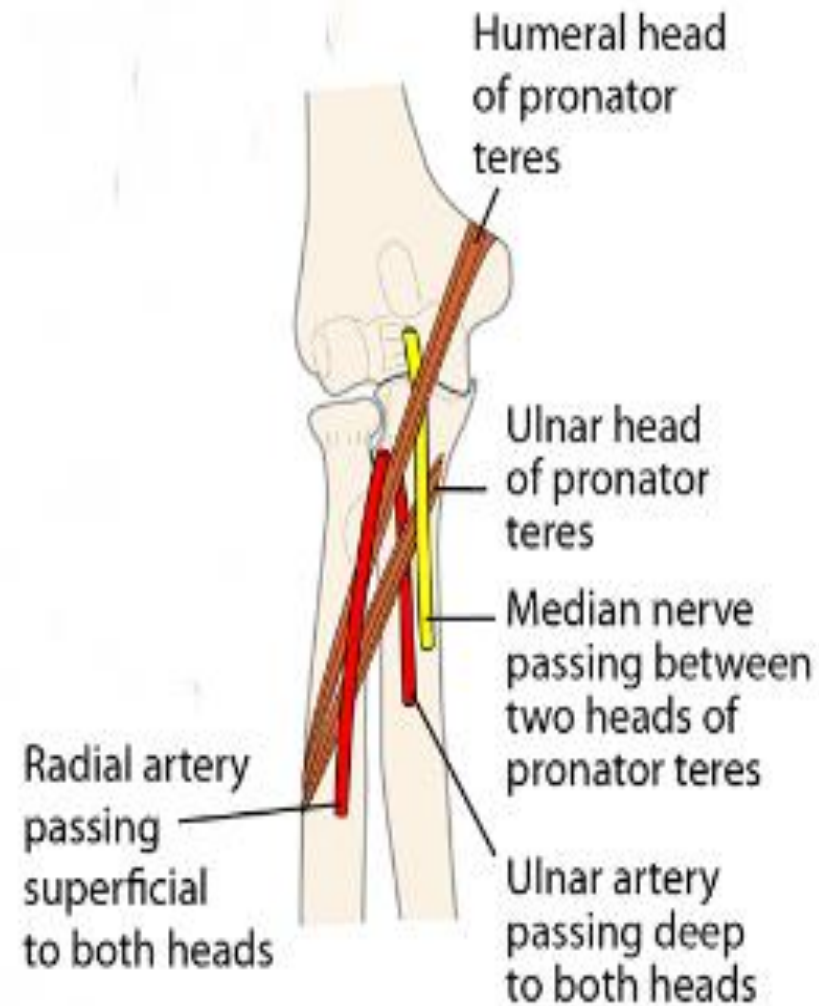
Anterior views

Relations of nerves & arteries in the elbow region



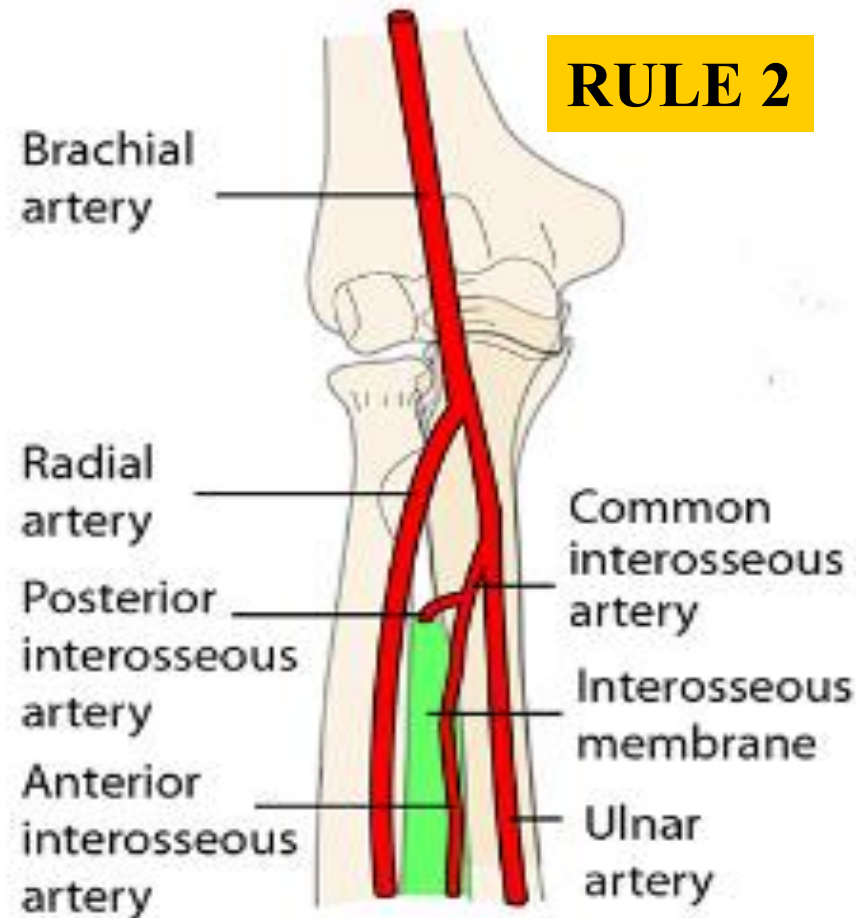
RULE 1

Nerves never pass between bones but pass around them.
Radial, median and ulnar nerves all pass between the two heads of muscles



Relations of the two heads of pronator teres

Relations of nerves & arteries in the elbow region



Arteries may pass between two bones as here where the posterior interosseous artery passes between ulna and radius

APPLIED ANATOMY

OF UL-I-

**Dr.Ahmed AL-
Baghdadi**

Humeral fractures

Dr. Ahmed AL-Baghdadi

Surgical Neck Fracture of Humerus

Case scenario: A 70-year-old woman presents after a **fall onto her outstretched hand** with pain and inability to lift her arm. Exam reveals ecchymosis over the proximal arm, crepitus on motion, and **intact radial pulse**

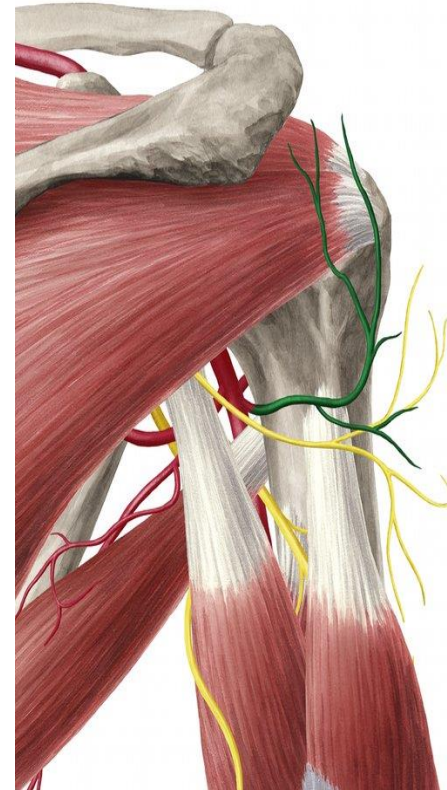


Surgical Neck Fracture of Humerus

Key Features:

- **High-risk population:** Elderly (osteoporosis), seizures, electrocution.
- **Critical structures at risk:**
 - Axillary nerve (runs 5 cm distal to acromion).
 - Posterior circumflex humeral artery (adjacent to nerve).
- **Deformity:** Proximal fragment abducted/externally rotated (pectoralis major pull)

Why "surgical": High risk for neurovascular injury (vs. "anatomical" neck).



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Midshaft Humeral Fracture

Case scenario: 35-year-old construction worker falls from scaffolding, landing on his right arm. He presents with **visible deformity at mid-arm**, **weak wrist extension**, and **numbness over the dorsum of the hand**.



Surgical Neck Fracture of Humerus

Key Features:

- **"Hanging arm" deformity**: Dependent distal fragment due to gravity.
- **Radial nerve jeopardy** : Wraps around spiral groove (10-14 cm distal to acromion)

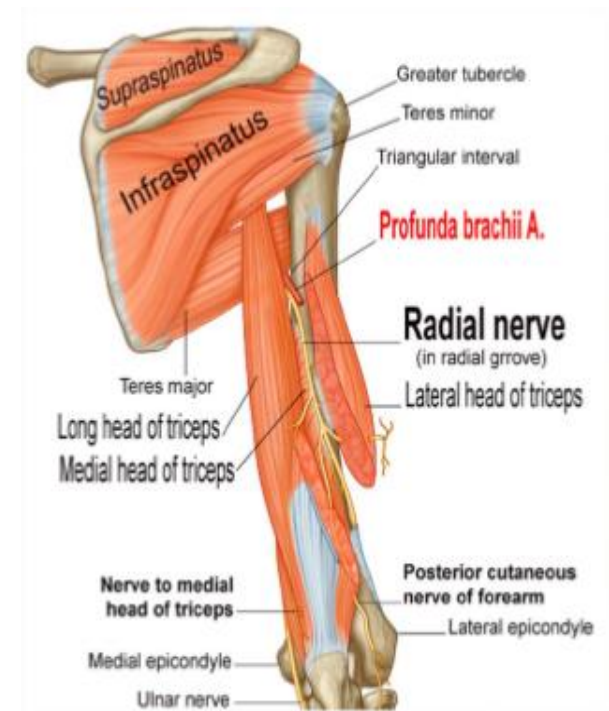
Anatomical features

1. Radial Nerve Course:

- **Proximal to fracture**: Innervates triceps (often spared in midshaft fractures).
- **At spiral groove**: Vulnerable to laceration/stretch (test: wrist/finger extension).
- **Distal to fracture**: Emerges lateral to brachioradialis (sensory: dorsal hand).

2. Muscle Forces & Displacement

- Proximal fragment: Pulled anteriorly by pectoralis major.
- Distal fragment: Pulled medially by brachialis/biceps.



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 in fractures of humeral shaft, the fractured segments displaced depends on site of fracture. Explain how is that?

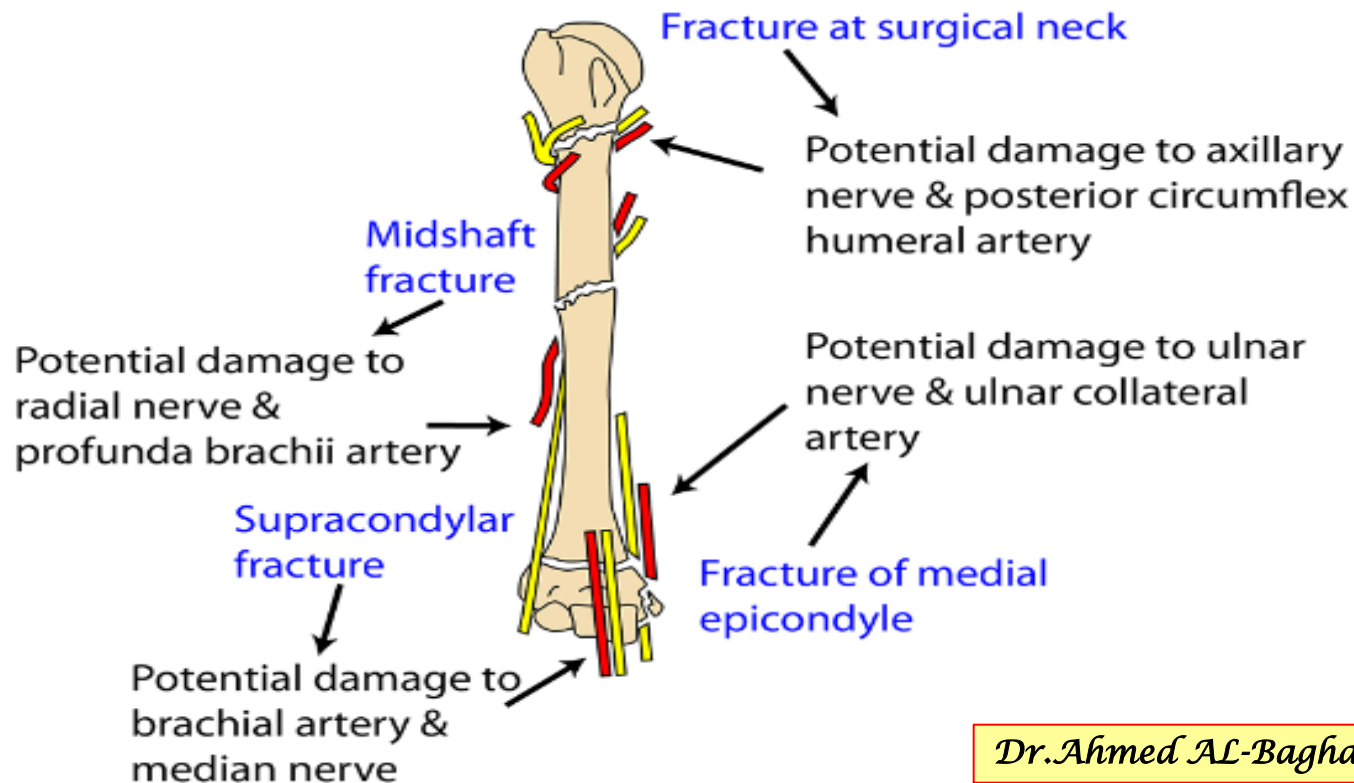
Site of fracture	Proximal segment	Distal segment
Above bicipital groove	Abducted by muscles attached to greater tuberosity(mainly supraspinatus)	Adducted by muscles attached to bicipital groove
Above the level of deltoid tuberosity	Adducted by muscles attached to bicipital groove	Abducted & pulled up by deltoid muscle
Below the level of deltoid tuberosity	Abducted by deltoid	Adducted & pulled up by biceps & triceps

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- ❑ What are the structures in direct contact with humerus and liable for injury?

ARTERIES AND NERVES LIABLE TO DAMAGE AT 4 SITES OF FRACTURE OF THE HUMERUS



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Pediatric Supracondylar Humerus Fractures

Case scenario: 6-year-old boy falls from monkey bars onto an **outstretched hand**, presenting with a swollen, **painful elbow held in extension**. Exam reveals no **radial pulse but warm, pink fingers**



Pediatric Supracondylar Humerus Fractures

Key Features:

- **High-risk population:** Ages 5-7 years (weak supracondylar area).
- **Critical structures at risk:**
 - Brachial artery (pulse loss in 5-15% of type III fractures).
 - Median/radial/ulnar nerves (anterior interosseous nerve most commonly injured).
- **"Snapping pop" sign:** Audible crack during injury suggests severe displacement

Pediatric Supracondylar Humerus Fractures

Anatomical features

Supracondylar Region:

- o Thin cortex: Weakest point between coronoid/olecranon fossae.
- o Growth plates: Distal humerus physis contributes 80% of elbow growth.

Neurovascular Relations:

- o Brachial artery: Anteromedial to fracture; pulse loss $\neq$ ischemia (collaterals may preserve hand perfusion).

Nerves:

- o Anterior interosseous (AIN): Test with "OK sign" weakness.
- o Radial: Test thumb/wrist extension.
- o Ulnar: Test finger abduction/intrinsic function.

Displacement Forces:

- o Extension-type (95%): Distal fragment displaces posteriorly (triceps pull).
- o Flexion-type (5%): Distal fragment displaces anteriorly (rare, higher neurovascular risk).

Pediatric Supracondylar Humerus Fractures

Anatomical features

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Compartment syndrome risk:

Volkman's ischemic contracture if missed vascular injury.

❑ What is Volkmann's ischemic contracture?

In supracondylar fracture

- There is displacement of proximal bone fragment forward & downward, compressing brachial a. & median n.
- When the collateral circulation cannot compensate for occlusion of the brachial artery leads to ischemia of flexor muscles of forearm & necrosis
- the muscles replaced by fibrous tissue leading to thinning & shortening of the muscle
- result in flexion deformity of wrist & fingers **(Volkmann's contracture)**



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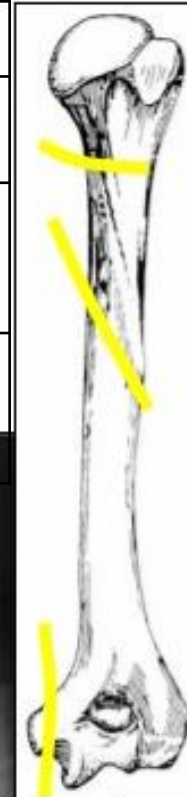
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Fractures of the humerus

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nerve	Site of contact	deformity
Axillary	Around surgical neck	Flat shoulder
Radial	Spiral groove	Wrist drop
ulnar	Behind medial epicondyle	Claw hand



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Radial Head Subluxation (Nursemaid's Elbow)

Case scenario: A 2-year-old girl presents with **sudden refusal to use her left arm** after her mother **pulled her up by the wrist**. The arm is held in **slight flexion and pronation** with **no swelling or deformity**



Radial Head Subluxation (Nursemaid's Elbow)

Key Features:

- **Classic mechanism:** Axial traction on a pronated forearm (e.g., lifting child by hands).
- **Pathognomonic finding:** No voluntary supination (child won't reach for toys with affected hand).
- **Age window:** Peak incidence 1-4 years (ligamentous laxity + immature radial head).

○ Anatomical features

Radial Head Development:

○ <5 years old: Radial head is immature narrow ovoid (not fully spherical) → easier subluxation.

○ Annular ligament: Thin, elastic and loose in toddlers; slips into radiocapitellar joint with traction.

Subluxation Mechanism:

○ Pronation + Traction → annular ligament slips over radial head → trapped in joint.

○ Reduction: Supination + flexion (lifts ligament out of joint).

